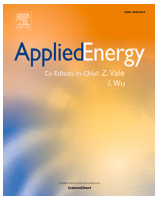
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PV-MM-diffusion: An end-to-end multi-modal diffusion model for ultra-short-term probabilistic photovoltaic forecasting

Jing Huang [a], Bopeng Shao [a], Yan Ke [a,b,*], Yuan Gao [c,d,**], Stefano Mazzoni [e]

[a] Information and Electronic Engineering, Zhejiang Gongshang University, Hangzhou, China
[b] YQ Future Energy Technology Co., Ltd, Hangzhou, China
[c] Institute of Advanced Study, Kyushu University, Fukuoka 816-8580, Japan
[d] The Center for Energy Systems Design (CESD), International Institute for Carbon-Neutral Energy Research (WPI-I2CNER), Kyushu University, Fukuoka, Japan
[e] Department of Industrial Engineering, University of Roma Tor Vergata, Rome, Italy

HIGHLIGHTS

- End-to-end diffusion model forecasts future sky images and PV power distributions.
- Cross-modal attention fuses visual and temporal features for enhanced predictability.
- Empty-frame placeholder strategy allows image-only, PV-only, or combined input.
- Achieves 18% CRPS and 40% Winkler score improvements on a 30 kW rooftop PV system.
- Provides multi-step probabilistic forecasts for risk-aware reserve allocation.

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ABSTRACT

Accurate ultra-short-term photovoltaic (PV) power forecasting is critical for the secure and efficient operation of power systems with high levels of renewable generation. Rapid cloud-cover variability, however, induces large and sudden fluctuations in PV output, which existing models struggle to capture and for which they provide limited measures of uncertainty. In this work, we propose PV-MM-Diffusion, an end-to-end multimodal diffusion model that jointly generates multiple future sky images and corresponding PV power distributions via two coupled denoising autoencoders. A cross-modal attention mechanism fuses complementary information from sky imagery and historical PV time series, and an empty-frame placeholder strategy allows the model to operate with image-only, PV-only, or combined inputs. Experiments on a 30 kW rooftop system show that PV-MM-Diffusion substantially improves probabilistic forecasting: it achieves a continuous ranked probability score (CRPS) of 2.63 kW (an 18 % reduction relative to the SkyGPT→U-Net baseline) and a Winkler score (WS) of 21.46 kW (a 40 % reduction). The model delivers tighter and more reliable prediction intervals, especially during extreme ramp events and rapid cloud transitions. These results demonstrate the promise of diffusion-based multimodal frameworks for flexible, uncertainty-aware PV integration in future low-carbon power systems.

1. Introduction

1.1. Research background

The global transition toward carbon neutrality and sustainable energy systems has driven rapid expansion of renewable generation, with photovoltaic (PV) power assuming a central role in future grids [1]. However, the inherent intermittency and high short-term variability of solar output—particularly due to rapid cloud movements on scales of seconds to 30 minutes—pose formidable challenges to secure and efficient grid operation [2]. Although flexible resources such as gas turbines can partially buffer these fluctuations, accurate ultra-short-term PV forecasting is indispensable for optimal dispatch, cost minimization, reliability enhancement, and deeper integration of renewables [3]. Inaccurate or delayed forecasts exacerbate scheduling complexity and incur additional economic and reliability risks across residential, commercial, and utility-scale PV applications [4,5]. As Song et al. point out, minute-level forecasts are essential for real-time inverter control and microgrid management [6]. Similarly, Xiong et al. emphasize that

* Corresponding author at: Information and Electronic Engineering, Zhejiang Gongshang University, Hangzhou, China.
** Corresponding author at: Institute of Advanced Study, Kyushu University, Fukuoka 816-8580, Japan.
Email addresses: yanke@mail.zjgsu.edu.cn (Y. Ke), gao.yuan.564@m.kyushu-u.ac.jp (Y. Gao).

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compensating for delayed storage responses requires accurate predictions of power trends several minutes in advance [7]. Precise forecasting of PV output is therefore essential to bolster the stability and reliability of grid-connected PV systems, optimize energy management and dispatch, and enable the effective integration of large-scale PV generation.

1.2. From deterministic to probabilistic photovoltaic forecasting

Over recent decades, solar energy forecasting has garnered increasing interest from researchers, grid operators, and market participants [8], as advances in artificial intelligence have popularized data-driven techniques such as artificial neural networks [9,10], support vector machines [11], and gradient boosting decision trees [12]. These methods relate PV output to inputs like meteorological observations [13], numerical weather predictions [14], sky camera sequences [15], and satellite imagery [16], and deliver point forecasts across horizons from ultra short term to medium and long term [17]. For example, Quan et al. [18] combined self-attention temporal convolutional networks with BiLSTM networks to boost ultra-short-term accuracy. Unlike physical simulators, data-driven models typically offer greater accuracy, flexibility, and ease of automation [19]. Moreover, they can adapt in real time as Luo et al.'s adaptive LSTM continually incorporates new data to refine day-ahead forecasts [20].

As solar photovoltaics account for a larger fraction of total electricity production, the consequences of forecasting errors on grid operations and costs have become increasingly significant [21]. Therefore, alongside point forecasts, it is now critical to characterize the uncertainty associated with PV output to manage risks in demand-side operations and electricity market participation [22]. Probabilistic forecasting addresses this need by estimating the complete probability distribution of future power generation [23], thereby delivering a richer portrayal of uncertainty and enabling stakeholders to make more informed, risk-aware decisions.

Probabilistic solar-power forecasting leverages various statistical and learning-based techniques to estimate the full distribution of future output. Gaussian processes (GPs) are widely used for their ability to capture uncertainty: for instance, van der Meer et al. applied GPs to residential load, PV generation, and net demand in a single household [24], while Sheng et al. introduced a weighted GP regression for short-term PV prediction [25]. Ensemble methods also play a key role: Cervone et al. combined artificial neural networks with ensemble sampling to produce deterministic and probabilistic 72-hour forecasts [26], and Alessandrini et al. employed an analog ensemble technique for PV output [27]. Quantile regression (QR) offers another route, directly estimating prediction intervals; Wen et al. compared bootstrap and QR approaches for PV uncertainty quantification [28]. Bayesian frameworks further enrich probabilistic forecasts: Panamtash et al. captured joint solar–temperature distributions with a copula-based Bayesian model [29].

Recent advances in deep learning have extended these methods. The DeepAR model exploits autoregressive recurrent networks trained across many related series to generate Monte Carlo–based predictive samples [30]. Hybrid LSTM–Bayesian neural network frameworks have been proposed for solar irradiance variability [31], and diffusion-based autoregressive architectures enable multivariate probabilistic time-series forecasting [32]. By combining sophisticated feature learning with rigorous uncertainty quantification, these deep-learning approaches yield rich, probability-consistent forecasts that better inform risk-aware decision-making.

1.3. Motivation, research gaps, and contributions

1.3.1. Research gaps and motivation for sky imagery

Probabilistic forecasting provides full predictive distributions or confidence intervals, which help grid operators allocate reserves, schedule flexible units, and manage economic and reliability risks when PV variability is high. However, current methods have three main limitations:

1. **Separate or single-step workflows.** Two-stage approaches such as SkyGPT to U-Net first predict future sky images and then perform PV regression. Single-step estimators predict only one time horizon at a time. Both workflows split visual and numerical data and cannot capture how uncertainty evolves over a 1 to 15 minute window.
2. **Weak interaction between modalities.** End-to-end CNNs [5,33], RNNs [34,35], and Transformer models [36] combine sky images with PV time series, but they often lag when clouds change suddenly. Their uncertainty quantification methods (for example, ensemble bootstrapping [37], bound prediction [38], or discrete binning [39]) remain basic and do not allow full information exchange between image and power data.
3. **Lack of unified generative models.** Diffusion models perform well on high-dimensional multimodal generation [40], yet they have not been applied in a single model to predict both sky images and PV output together.

Ground-based sky cameras capture cloud movement at high spatiotemporal resolution. Cloud motion is the main cause of rapid PV power changes and cannot be reproduced by time-series data alone [41]. Using sky imagery addresses the gaps mentioned above:

- **Adds physical context.** Fisheye images capture cloud motion every few seconds, giving clear information about upcoming PV power increases or drops.
- **Supports true multimodal fusion.** Modeling visual and numerical data together allows the model to predict cloud development and the resulting PV distribution at the same time.
- **Solves two-stage shortcomings.** Previous work showed that generative sky-image prediction helps uncertainty quantification [15], but splitting the process into two stages prevents consistent propagation of uncertainty. A unified diffusion model fixes this issue.

1.3.2. Contributions

To overcome these challenges, we propose **PV-MM-Diffusion**, the first end-to-end diffusion model that handles both sky images and PV data for ultra-short-term forecasting. Our contributions are:

1. **Unified multimodal diffusion.** A single diffusion network that generates future sky images and the corresponding PV power distributions together, allowing full interaction between image and power data.
2. **Multi-step probabilistic forecasts.** Treating the task as a conditional diffusion process, the model directly outputs predictive distributions for each minute up to 15 minutes ahead, improving uncertainty calibration.
3. **Cross-modal attention fusion.** A bidirectional attention module aligns features from images and time series, making the model more responsive to sudden cloud-induced power changes.
4. **Flexible framework extensibility.** The proposed architecture is inherently compatible with diverse input modalities and spatial scales. Although this manuscript specifically demonstrates its effectiveness on a single rooftop PV system, the underlying multimodal diffusion formulation is designed to be naturally extensible to broader sensing data (e.g., satellite imagery) and larger-scale PV networks in future studies.
5. **Empirical validation.** On a 30 kW rooftop PV system, PV-MM-Diffusion outperforms state-of-the-art benchmarks in CRPS and Winkler score (WS), especially during extreme ramp events.

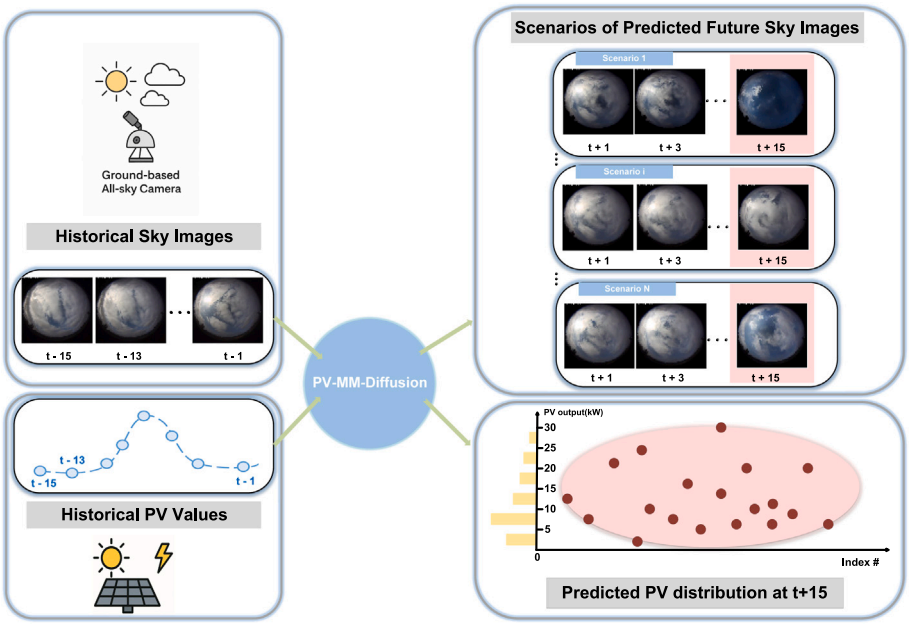


Fig. 1. Overview of the proposed multimodal probabilistic PV forecasting framework. Specifically, the framework first takes as input a sequence of historical sky images collected over the past 16 minutes (acquired every 2 minutes) and the corresponding historical photovoltaic generation data. Subsequently, at the same temporal resolution, it synchronously forecasts the future sky image sequence and the photovoltaic output sequence over the next 15 minutes. The predictions of both the sky image sequence and the photovoltaic output sequence are generated in an iterative manner. Finally, the full set of photovoltaic power samples predicted at time $t + 15$ is treated as the probability distribution of the photovoltaic output, thereby completing the probabilistic forecast of the photovoltaic output.

2. Methodology

This study focuses on ultra-short-term probabilistic photovoltaic (PV) forecasting, emphasizing a 15-minute-ahead PV power interval prediction. To achieve this goal, we propose an end-to-end multi-modal PV forecasting diffusion model, which can simultaneously generate various possible future sky images and their corresponding PV output probability distributions. Fig. 1 illustrates a schematic diagram of the proposed multi-modal probabilistic PV forecasting framework.

To compare the predictive performance differences arising from different input and output configurations, based on the aforementioned end-to-end multi-modal forecasting diffusion model, this study further designs an "empty-frame placeholder" scheme. This scheme enables the model to remain compatible with multiple data modes in both the training and forecasting phases without changing the network architecture. Three input-output modes are supported:

1. **PV-MM-Diffusion ("SI PV → SI PV"):** Both historical sky images and historical PV sequences are input, and future sky images as well as PV sequences are output.
2. **"SI → SI PV":** Only historical sky images are input, and future sky images and PV sequences are output.
3. **"PV → PV":** Only historical PV sequences are input, and future PV sequences are output.

First, Section 2.1 provides a high-level intuition of the proposed multimodal diffusion process and a concise summary of how uncertainty is jointly propagated across sky images and PV sequences. Then, Section 2.2 briefly overviews the fundamentals of diffusion models. Then, Section 2.3 introduces the construction of PV-MM-Diffusion based on DDPM, which extends the original diffusion model to achieve semantically consistent multimodal forecasting. Section 2.4 describes in detail the Coupled U-Net architecture and the cross-modal attention module used for synchronous modeling of PV and sky image sequences. Section 2.5 explains the specific implementation principle of the "empty-frame placeholder" scheme. Finally, Section 2.6 defines the evaluation metrics used in this study to assess the predictive performance of the proposed model.

2.1. High-level intuition and notation

Goal (what we model). Given historical sky images and PV observations, our goal is to learn a *joint* predictive distribution over the next 15 minutes:

$$(\hat{S}^{8:15}, \hat{P}^{8:15}) \sim p_\theta\left(S^{8:15}, P^{8:15} \mid S^{0:7}, P^{0:7}\right), \tag{1}$$

where $S^{0:7}$ and $P^{0:7}$ are the historical context, and $S^{8:15}$ and $P^{8:15}$ are the aligned future sequences to be generated.

High-level intuition (multimodal diffusion). PV-MM-Diffusion follows the standard diffusion idea: (i) we *corrupt* the unknown future part into Gaussian noise via a forward process; (ii) we learn a *conditional denoiser* that iteratively removes noise to recover plausible futures. Crucially, we denoise *sky images and PV sequences together* using a coupled U-Net with cross-modal attention, so that cloud-motion uncertainty and PV ramp uncertainty are learned and resolved in a single end-to-end process.

Concise summary: how uncertainty is jointly propagated across modalities. At inference, we start from random noise for both future modalities and run the reverse denoising steps. Each random seed produces one *paired* future sample $(\hat{S}^{8:15}, \hat{P}^{8:15})$. Because the two modalities are denoised by the *same* coupled network and exchange features via cross-modal attention at every denoising step, ambiguity in future clouds (multiple plausible sky evolutions) is *synchronously* reflected in multiple plausible PV trajectories. Repeating sampling yields an empirical predictive distribution for PV at each horizon, while maintaining consistency with the generated sky scenarios.

Notation (to avoid confusion). We explicitly distinguish the *forecast timestamp* from the *diffusion step*:

- Real-world forecasting time is denoted by t (e.g., t+15 minutes).

- Diffusion step is denoted by $\tau \in \{0, 1, \dots, T\}$ (to avoid overloading t).
- Superscripts $0{:}7$ and $8{:}15$ denote *frame indices* of the context and target sub-sequences in a concatenated 16-step sample (8 context frames + 8 future frames), rather than "minutes". The physical lead time is determined by the sampling interval Δt (2 minutes in this dataset), and the target range $8{:}15$ corresponds to the future horizons within the 15-minute forecasting window.
- We reserve T for the number of diffusion steps (e.g., T=1000), and use L for the sequence length in the network input (here L=16).

2.2. Preliminaries of vanilla diffusion

Diffusion-based models [42,43] refer to a class of generative algorithms that first transform a given data distribution x into unstructured noise (Gaussian noise in practice), and then learn to recover the original data distribution through the reverse process (i.e., the aforementioned forward process). In the original Denoising Diffusion Probabilistic Models (DDPMs) [42], the forward process is carried out over T discrete time steps. Let x_0 be a sample drawn from the data distribution X, and let x_T be a sample drawn from a Gaussian noise distribution, which is independent of x_0 in a Markov process. To avoid confusion with the real forecasting time t (e.g., $t+15$ minutes), we use τ to index diffusion steps throughout Sections 2.2–2.4. This forward process is formulated as:

$$q(x_\tau \mid x_{\tau-1}) = \mathcal{N}\left(x_\tau; \sqrt{1-\beta_\tau}\, x_{\tau-1}, \beta_\tau \mathbf{I}\right) \tag{2}$$

$$q(x_{1:T} \mid x_0) = \prod_{\tau=1}^{T} q\left(x_\tau \mid x_{\tau-1}\right) \tag{3}$$

where $\tau \in [1, T]$, and $\beta_1, \dots, \beta_T$ is a predefined variance schedule. Following previous work [42], we adopt a linear noise schedule.

For later use (e.g., Algorithm 1), the marginal distribution at any diffusion step admits a closed form:

$$q(x_\tau \mid x_0) = \mathcal{N}\Big(x_\tau; \sqrt{\bar{\alpha}_\tau}\, x_0, (1-\bar{\alpha}_\tau)\mathbf{I}\Big), \quad \alpha_\tau = 1-\beta_\tau, \quad \bar{\alpha}_\tau = \prod_{s=1}^{\tau} \alpha_s. \tag{4}$$

To recover the original image, one can simplify the learning of the reverse process into training a model θ to fit $p_\theta(x_{\tau-1} \mid x_\tau)$, so that for all given τ and x_τ, the model can closely approximate $q(x_{\tau-1} \mid x_\tau, x_0)$. Consequently, the reverse process can be written in Eq. (5), and x_0 can be recovered from the probability density $p(x_T)$ by Eq. (6). Specifically:

$$p_\theta(x_{\tau-1} \mid x_\tau) = \mathcal{N}\left(x_{\tau-1}; \mu_\theta(x_\tau, \tau), \Sigma_\theta(x_\tau, \tau)\right) \tag{5}$$

$$p_\theta(x_{0:T}) = p(x_T) \prod_{\tau=1}^{T} p_\theta\left(x_{\tau-1} \mid x_\tau\right) \tag{6}$$

Here, μ_θ denotes the Gaussian mean predicted by model θ, which ultimately yields x_0. In practice, since variance prediction provides only marginal improvements [44,45], we omit it, and this part will also be omitted in the following text.

2.3. PV-multi-modal diffusion models

Based on Section 2.2, we extend DDPM to a *joint* multimodal setting. We aim to generate two aligned future modalities within the same diffusion trajectory: the future PV sequence $P_\tau^{8:15}$ and the future sky image sequence $S_\tau^{8:15}$, conditioned on historical observations $P^{0:7}$ and $S^{0:7}$. Here, the superscripts denote temporal ranges (historical vs. future), while the subscript τ denotes the diffusion step (not the real time).

As shown in Fig. 2, in contrast to conventional diffusion models that solely predict a single modality, our objective is to predict two modalities within the same diffusion process, ensuring consistency between these modalities. Meanwhile, the historical PV sequence $P^{0:7}$ and the historical sky image sequence $S^{0:7}$ are used as conditional inputs.

Given paired data (P, S) from a one-dimensional PV sequence set (including both conditional and predictive parts) PV ($P \in$ PV) and a three-dimensional sky image sequence set (including both conditional and predictive parts) SI ($S \in$ SI), and considering that each modality follows a distinct distribution, the forward process can be viewed as independent across modalities. Taking the PV sequence P as an example, its future portion in the forward process at step τ is defined as follows:

$$q\left(P_\tau^{8:15} \mid P_{\tau-1}^{8:15}\right) = \mathcal{N}_P\Big(P_\tau^{8:15} \;\Big|\; \sqrt{1-\beta_\tau}\, P_{\tau-1}^{8:15}, \beta_\tau \mathbf{I}\Big), \quad \tau \in [1, T] \tag{7}$$

During the forward diffusion, the historical context $P^{0:7}$ is kept fixed, and noise is injected only into the future segment $P^{8:15}$ to be generated. The sky-image future segment $S^{8:15}$ follows the same schedule $\{\beta_\tau\}_{\tau=1}^{T}$ (omitted for brevity). Using Eq. (4), we can directly obtain $P_\tau^{8:15}$ and $S_\tau^{8:15}$ at any diffusion step τ in closed form.

Unlike approaches that independently model the future PV and future sky image sequences in the forward process, our reverse denoising stage must consider the overall correlation between the two modalities while also maintaining the relationships between conditional and predictive components within each modality. Therefore, we propose a unified model θ_{PS} that takes both modalities (including their conditional and predictive parts) as inputs. For any given diffusion step τ, the reverse process of the future PV sequence $P_{\tau-1}^{8:15}$ in the PV domain is:

$$\begin{aligned} &p_{\theta_{PS}}\Big(P_{\tau-1}^{8:15} \;\Big|\; \left(P^{0:7}, P_\tau^{8:15}, S^{0:7}, S_\tau^{8:15}\right)\Big) \\ &= \mathcal{N}\Big(P_{\tau-1}^{8:15} \;\Big|\; \mu_{\theta_{PS}}^{P}\left(P^{0:7}, P_\tau^{8:15}, S^{0:7}, S_\tau^{8:15}, \tau\right)\Big) \end{aligned} \tag{8}$$

The reverse denoising for $S_\tau^{8:15}$ is defined analogously. Importantly, both modalities are denoised by the coupled network θ_{PS} at each step, which enables step-by-step cross-modal uncertainty propagation. To optimize the entire network, we employ the ε-prediction strategy, whereby the model directly regresses the noise added to the "future" portion.

Algorithm 1 "Empty-Frame Placeholder" based multimodal diffusion training procedure.

Require: Training dataset D, where each sample contains a PV sequence P and an image sequence S; multimodal diffusion model Model
Output: The trained parameters of Model

repeat
  Batch ← randomly sampled mini-batch from D
  for each (P, S) in **Batch do**
    $P_0^{0:7}$, $P_0^{8:15}$ ← $P[0{:}7]$, $P[8{:}15]$
    $S_0^{0:7}$, $S_0^{8:15}$ ← $S[0{:}7]$, $S[8{:}15]$
    mode_rand ← random.random()
    if *mode_rand* $< \frac{1}{3}$ **then** ▷ PV-MM-Diffusion ("SI PV → SI PV")
      `// Keep the real values of` $P_0^{0:7}$ `and` $S_0^{0:7}$
    else if *mode_rand* $< \frac{2}{3}$ **then** ▷ "SI → SI PV"
      $P_0^{0:7} \leftarrow 0$
    else ▷ "PV → PV"
      $S_0^{0:7} \leftarrow 0$
    end if
    $\tau \leftarrow$ Uniform$\{1, \dots, T\}$ ▷ sample a diffusion step
    $P_\tau^{8:15} = \sqrt{\bar{\alpha}_\tau}\, P_0^{8:15} + \sqrt{1-\bar{\alpha}_\tau}\, \varepsilon_P \quad \leftarrow \quad \varepsilon_P \sim \mathcal{N}_{P^{8:15}}(0, I)$
    $S_\tau^{8:15} = \sqrt{\bar{\alpha}_\tau}\, S_0^{8:15} + \sqrt{1-\bar{\alpha}_\tau}\, \varepsilon_S \quad \leftarrow \quad \varepsilon_S \sim \mathcal{N}_{S^{8:15}}(0, I)$
    $\varepsilon_\theta^P, \varepsilon_\theta^S \leftarrow \text{Model}\left(P_0^{0:7}, P_\tau^{8:15}, S_0^{0:7}, S_\tau^{8:15}, \tau\right)$
    if *mode_rand* $< \frac{2}{3}$ **then**
      Take gradient step on $\nabla_\theta \left\|\varepsilon_\theta^P - \varepsilon_P\right\|_2^2$ and $\nabla_\theta \left\|\varepsilon_\theta^S - \varepsilon_S\right\|_2^2$
    else
      Take gradient step on $\nabla_\theta \left\|\varepsilon_\theta^P - \varepsilon_P\right\|_2^2$
    end if
  end for
until convergence

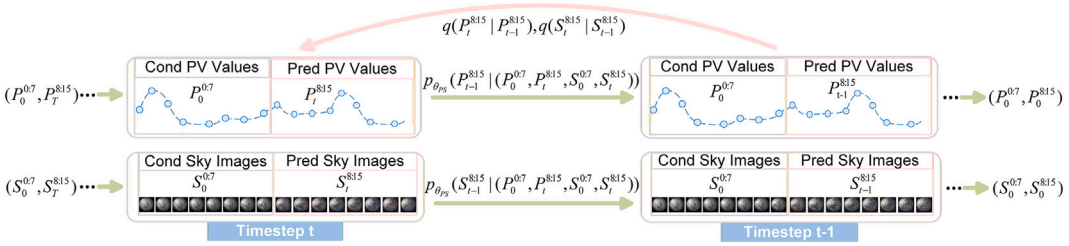


Fig. 2. Multimodal diffusion-denoising process for joint forecasting of sky images (SI) and PV power. The forward diffusion (pink arrow) progressively Adds Gaussian noise to the *future* segments ($S^{8:15}$, $P^{8:15}$) over diffusion steps $\tau = 1, \ldots, T$, while keeping the historical context ($S^{0:7}$, $P^{0:7}$) fixed. The reverse denoising (green arrows) uses the coupled model θ_{PS} with cross-modal feature exchange to generate temporally aligned future SI and PV samples.

Since the conditional parts do not require denoising, the loss function only applies to the predictive parts, defined as:

$$\ell_{\theta_{PS}} = \mathbb{E}_{\tau\sim\mathrm{Unif}\{1,\ldots,T\}}\mathbb{E}_{\varepsilon_P,\varepsilon_S\sim\mathcal{N}(0,I)}\left[\lambda(\tau)\left(\left\|\varepsilon_\theta^P(\cdot)-\varepsilon_P\right\|_2^2+\left\|\varepsilon_\theta^S(\cdot)-\varepsilon_S\right\|_2^2\right)\right] \quad (9)$$

where $(\cdot)$ abbreviates the common conditioning tuple $(P^{0:7}, P_\tau^{8:15}, S^{0:7}, S_\tau^{8:15}, \tau)$. Through this multi-modal diffusion framework, the proposed unified model θ_{PS} is capable of more comprehensively capturing the semantic correlations between PV and sky image sequences in a multi-modal feature space. Meanwhile, it can leverage the historical conditional information of both modalities to infer their future evolutions. During the prediction phase, the "future" portions of both modalities can complement and inform each other, thereby mitigating uncertainties arising from the limited information of a single modality, ultimately leading to more accurate predictions.

2.4. Coupled U-Net for future PV sequence–future sky image sequence denoising

We propose a joint U-Net architecture tailored for PV sequences and sky image sequences, referred to as the Coupled U-Net (see Fig. 3(a)). By incorporating the conditional prior (historical PV sequences and historical sky image sequences) introduced in Section 2.3, this architecture is able to more accurately predict future PV power outputs and future sky images during the denoising process.

We use L to denote the temporal length of the concatenated sequence input to the network (here $L = 16$). Specifically, we denote the inputs of the PV and sky image sequences as a pair of tensors $(P, S) \in (\mathrm{PV}, \mathrm{SI})$, which include both the conditional and predictive parts. Here, the PV input is $P \in \mathbb{R}^{C\times L}$ (where C is the number of channels and L is the temporal dimension), while the sky image sequence input is $S \in \mathbb{R}^{L\times C\times H\times W}$ (where L is the temporal dimension, C is the number of channels, and H and W denote the height and width of the image, respectively).

Efficient multimodal module. As illustrated in Fig. 3(b), in order to capture both temporal and spatial information while reducing computational overhead, we draw upon the approach proposed in MM-Diffusion [46] for video processing when designing the subnetwork for sky image sequences. Concretely, we separate spatial convolutions from temporal convolutions: first, 2D convolutions are employed to encode the spatial resolution, and then 1D convolutions are stacked along the temporal dimension. This strategy avoids the heavy computational cost that would result from directly using 3D convolutions. Meanwhile, to capture variations in cloud patterns across different dimensions more effectively, the sky image sequence subnetwork also incorporates two types of attention modules: temporal attention (1D) and spatial attention (2D). These modules facilitate more efficient modeling of the cloud dynamics over time and space. For the PV subnetwork, since there are only 16 time steps and each time step contains a single feature value, we adopt a strategy of "1D convolution + temporal attention", progressively increasing the number of channels as the network depth grows without performing downsampling along the temporal axis. This ensures maximal retention of key information in the PV sequence during the encoder phase. Subsequently, alignment with the sky image sequence subnetwork is achieved at the multimodal attention layer. In the decoder phase, spatiotemporal information is progressively recovered through inverse operations at each layer. By integrating the designs described above, the Coupled U-Net can simultaneously focus on the feature representations of both PV and sky image modalities during the denoising process, in combination with the diffusion model framework introduced in Section 2.2. Through the inclusion of the conditional prior described in Section 2.3, this model not only optimizes the deep semantic correlations between PV and sky image sequences in the multimodal feature space, but also infers the respective predictive components based on the conditional segments of each modality. Moreover, it facilitates interactive learning between these predictive components, ultimately leading to a marked improvement in predictive accuracy during the final inference stage.

Random-shift based PV-multi-modal attention. When aligning and performing joint learning on the features from the PV sequences and sky image sequences, the most straightforward approach would be to apply a global cross-modal attention to both modalities. However, given the large spatial dimension of sky images and the temporal dimension of PV data, the computational complexity of naive global attention can be as high as $O\big((L\times H\times W)\times L\big)$. Moreover, both the sky image sequences and PV sequences exhibit redundant information across temporal and spatial dimensions, and not all parts of these sequences are necessary for cross-modal attention calculations. To address this issue, we draw on the RS-MMA approach from [46] and tailor it to the features of PV and sky image sequences, forming what we call Random-Shift based PV-Multi-Modal Attention (RS-PV-MMA), as illustrated in Fig. 3(c). Concretely, for the lth layer of the Coupled U-Net, the output feature dimensions are (H^l, W^l, C^l, L). The sky-image tensor S with L frames can be viewed as a sequence of patches of size $[L \times H^l \times W^l]$, while the PV tensor P is represented as a temporal feature sequence of size $[C^l \times L]$. Cross-modal attention is performed via the following two steps:

Step I: We define a temporal window length L_w that is much smaller than the total number of frames L, and randomly sample an offset R from the interval $[0, L - L_w]$. When calculating the attention from the PV sequence to the sky image sequence, the ith PV segment P_i is paired with the corresponding sky image sequence segment S_j from frame f_s to frame f_e, where

$$f_s = (i + R) \bmod L, \qquad f_e = (i + R + L_w) \bmod L.$$

This setup enables each PV value to attend to a distinct local subset of the sky image sequence, thus reducing unnecessary cross-modal attention computations.

Step II: For the cross-modal attention between the PV sequence segment P_i and the sampled sky image segment $S_j = S_{f_s:f_e}$, we have:

$$\mathrm{MMA}(P_i, S_j) = \mathrm{softmax}\left(\frac{Q_i^P (K_j^S)^T}{\sqrt{d_k}}\right) V_j^S, \quad (10)$$

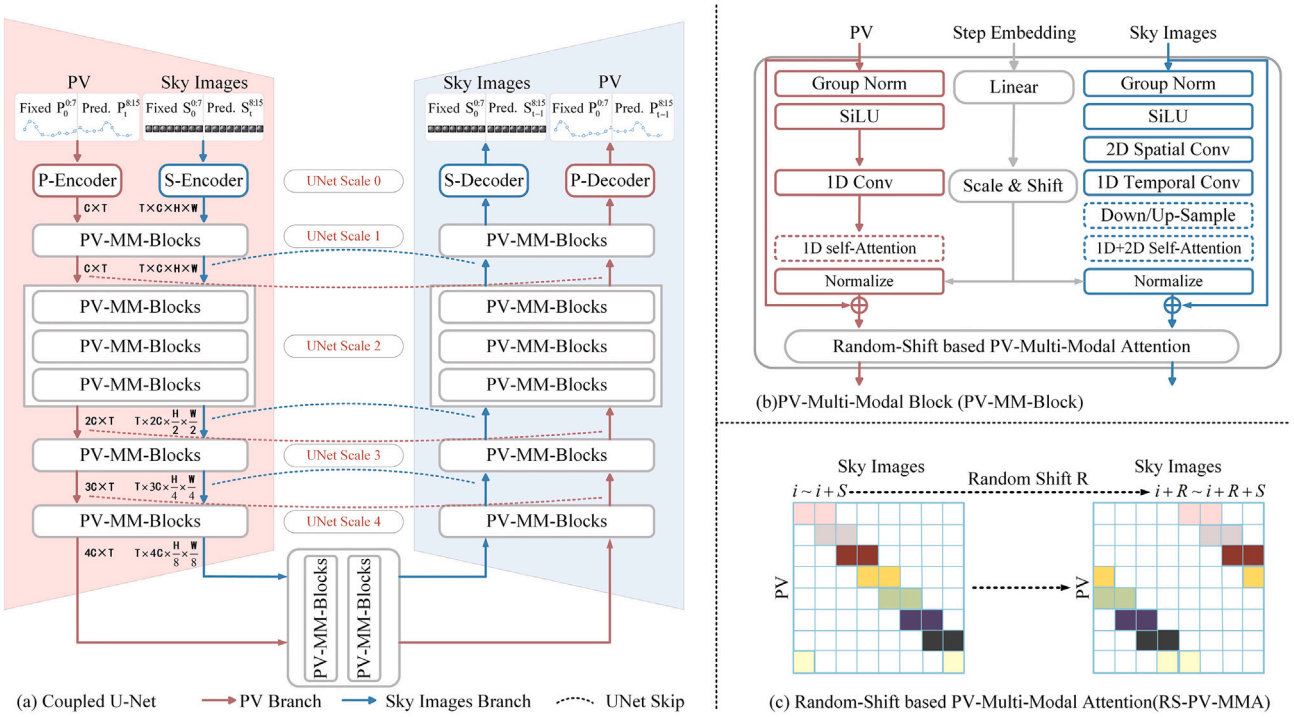


Fig. 3. Architecture of PV-MM-Diffusion. (a) coupled U-Net denoiser θ_{PS} with two branches for PV sequences (1D) and sky-image sequences (2D + temporal), exchanging information Through multi-scale skip connections and cross-modal attention. (b) PV-MM-Block: 1D convolution and temporal self-attention for PV feature extraction; spatial 2D convolution, temporal 1D convolution, and attention for SI to capture spatiotemporal cloud dynamics. (c) Random-Shift PV multi-modal attention (RS-PV-MMA): a local-window cross-attention module (window size L_w) that reduces cross-modal complexity while preserving temporal alignment between SI patches and PV tokens.

$$K_j^S = \mathrm{Linear}\big(\mathrm{flatten}(S_j)\big). \tag{11}$$

The cross-modal attention from the sky image sequence to the PV sequence, $\mathrm{MMA}(S_j, P_i)$, is symmetric and thus omitted here for brevity.

By using local windows, the complexity is reduced from $O\big((L\times H\times W)\times L\big)$ to approximately $O\big((L_w\times H\times W)\times L_w\big)$, substantially lowering the cost of cross-modal computation while preserving local temporal correspondences. Because cross-modal attention facilitates bidirectional interaction between the PV and sky image sequence features, the U-Net's top layers can leverage a smaller window size L_w to capture more detailed feature correspondences. In contrast, larger window sizes are employed in the deeper (bottom) layers of the U-Net to capture higher-level semantic alignments.

2.5. "Empty-frame placeholder" input scheme

In order to compare the impact of different input and output configurations on predictive performance and to maintain consistency of the network structure during both training and inference, we propose an "empty-frame placeholder" scheme. This scheme enables a single multimodal predictive diffusion model to be compatible with multiple data modalities. Specifically, the scheme supports the following three input-output forms:

1. **PV-MM-Diffusion ("SI PV → SI PV")**: Historical sky image sequences and photovoltaic (PV) sequences are simultaneously input, and the future sky image sequences and PV sequences are output.
2. **"SI → SI PV"**: Only historical sky image sequences are input, while the model outputs both future sky image sequences and future PV sequences.
3. **"PV → PV"**: Only historical PV sequences are input, while the model outputs future PV sequences.

In particular, for those modalities not required at a given time, a zero tensor is used as a placeholder to ensure that the dimensionality and temporal length of the network input remain consistent during both training and inference. By fusing and modeling the features under different conditions, the proposed scheme effectively accommodates diverse data patterns.

Basic principle. As mentioned above, taking the PV sequence P as an example, an ideal multimodal model learns the joint conditional distribution $p\big(P_{\tau-1}^{8:15}\,\big|\,(P_0^{0:7}, P_\tau^{8:15}, S_0^{0:7}, S_\tau^{8:15})\big)$, where τ denotes the diffusion step (cf. Section 2.1).

In practical operations, the "empty-frame placeholder" approach is adopted to handle missing modalities: if only the historical sky image sequences are provided, we set the historical PV sequence $P_0^{0:7} = 0$ to be a zero tensor; if only the historical PV sequences are provided, we set the historical sky image sequence $S_0^{0:7} = 0$ to be a zero tensor; if both modalities are provided, we use their actual values. Through this placeholder operation, the network can uniformly process the three scenarios (simultaneous input of historical sky image sequences and PV sequences, input of only historical PV sequences, and input of only historical sky image sequences) during both training and inference. This obviates the need for multiple independent models, thereby enhancing flexibility and generality, reducing model management complexity, and ensuring consistent prediction results.

Loss function design. During training, to jointly learn the three input–output modes within a single network, we adopt a *mixed-mode* objective with non-negative mixture weights $\boldsymbol{w} = (w_{\mathrm{SP}}, w_{\mathrm{SI}}, w_{\mathrm{PV}})$, where $w_{\mathrm{SP}} + w_{\mathrm{SI}} + w_{\mathrm{PV}} = 1$. Here, w_{SP}, w_{SI}, and w_{PV} correspond to the modes "SI PV→SI PV", "SI→SI PV", and "PV→PV", respectively. Unless otherwise stated, we use the balanced setting $w_{\mathrm{SP}} = w_{\mathrm{SI}} = w_{\mathrm{PV}} = \frac{1}{3}$. The sensitivity to $\boldsymbol{w}$ is reported in Section 4.3 (Table 4).

$$\ell_{\theta_{PS}} = \mathbb{E}_{\tau\sim\mathrm{Unif}\{1,\dots,T\}}\mathbb{E}_{\epsilon_P,\epsilon_S\sim\mathcal{N}(0,I)}\Big[w_{\mathrm{SP}}\Big(\big\|\epsilon_\theta^P(C_{\mathrm{SP}})-\epsilon_P\big\|_2^2+\big\|\epsilon_\theta^S(C_{\mathrm{SP}})-\epsilon_S\big\|_2^2\Big) + w_{\mathrm{SI}}\Big(\big\|\epsilon_\theta^P(C_{\mathrm{SI}})-\epsilon_P\big\|_2^2+\big\|\epsilon_\theta^S(C_{\mathrm{SI}})-\epsilon_S\big\|_2^2\Big) + w_{\mathrm{PV}}\big\|\epsilon_\theta^P(C_{\mathrm{PV}})-\epsilon_P\big\|_2^2\Big], \tag{12}$$

where $C_{\mathrm{SP}} = (P_0^{0:7}, P_\tau^{8:15}, S_0^{0:7}, S_\tau^{8:15}, \tau)$, $C_{\mathrm{SI}} = (0, P_\tau^{8:15}, S_0^{0:7}, S_\tau^{8:15}, \tau)$, $C_{\mathrm{PV}} = (P_0^{0:7}, P_\tau^{8:15}, 0, S_\tau^{8:15}, \tau)$.

In Eq. (12), the three terms correspond to PV-MM-Diffusion ("SI PV → SI PV"), "SI → SI PV", and "PV → PV", respectively. Note that in the ("PV → PV") scenario, the loss for the predicted sky-image sequence is not computed.

Training and inference process. To illustrate more intuitively how the "empty-frame placeholder" scheme is executed in a multimodal diffusion model, Algorithms 1 and 2 provide detailed pseudocode for training and inference, respectively. These algorithms explain how to perform placeholder operations under different input modes, how the mixed loss is adopted, and how the future PV sequences and sky image sequences are recovered via reverse diffusion steps.

2.6. Performance evaluation metrics

Video prediction. The predicted image frames are assessed both qualitatively, based on human visual inspection, and quantitatively, using the widely adopted metric of VGG cosine similarity (VGG CS) [47] for benchmarking video-prediction models:

$$\mathrm{VGG\ CS} = \frac{1}{N}\sum_k^N\left(\frac{\boldsymbol{v}_k\cdot\hat{\boldsymbol{v}}_k}{\|\boldsymbol{v}_k\|\|\hat{\boldsymbol{v}}_k\|}\right) \tag{13}$$

Algorithm 2 "Empty-Frame Placeholder" based multimodal diffusion inference procedure.

Require: Test set D_{test}, where each sample contains a PV sequence P and an image sequence S; trained multimodal diffusion model Model; total reverse diffusion steps T (using DPMSolver++)
Output: Predicted PV sequence $(\hat{P}^{8:15})$ and image sequence $(\hat{S}^{8:15})$
for each (P, S) in D_{test} **do**
  1) Split data and place placeholders
  $P_0^{0:7}, _ \leftarrow P[0{:}7],\ P[8{:}15]$
  $S_0^{0:7}, _ \leftarrow S[0{:}7],\ S[8{:}15]$
  if only sky image sequence **then**
    $P_0^{0:7} \leftarrow 0$
  end if
  if only PV sequence **then**
    $S_0^{0:7} \leftarrow 0$
  end if
  2) Initialize the latter part with Gaussian noise
  $P_T^{8:15} \leftarrow \mathcal{N}(0, I)$
  $S_T^{8:15} \leftarrow \mathcal{N}(0, I)$
  3) Reverse diffusion (from $\tau = T$ down to 1) using DPMSolver++
  for τ in range$(T, 0, -1)$ **do**
    $\epsilon_\theta^P,\ \epsilon_\theta^S \leftarrow \mathrm{Model}\big(P_0^{0:7}, P_\tau^{8:15}, S_0^{0:7}, S_\tau^{8:15}, \tau\big)$
    `// Apply DPMSolver++ adaptive step for reverse update`
    $P_{\tau-1}^{8:15},\ S_{\tau-1}^{8:15} \leftarrow \mathrm{DPMSolverPlusPlusUpdate}\big(P_\tau^{8:15}, S_\tau^{8:15}, \epsilon_\theta^P, \epsilon_\theta^S, \tau\big)$
  end for
  4) Final prediction
  $\hat{P}^{8:15} \leftarrow P_0^{8:15},\quad \hat{S}^{8:15} \leftarrow S_0^{8:15}$
end for

where $\boldsymbol{v}_k = \mathrm{VGG}(I_k)$ and $\hat{\boldsymbol{v}}_k = \mathrm{VGG}(\hat{I}_k)$. The operator *VGG* denotes feature extraction using the VGG-16 network [48]. We employ the VGG-16 model pre-trained on ImageNet and distributed via the TensorFlow–Keras API. The feature vectors correspond to the output of the penultimate max-pooling layer. Because VGG-16 expects an input resolution of 224×224, the original 64×64 frames are resized with the Python OpenCV library before computing VGG CS. VGG CS has been shown to correlate more strongly with human perceptual judgments than pixelwise metrics [47]. This quantitative indicator is primarily used to evaluate deterministic forecasts. For stochastic video-prediction models, we generate 30 candidate futures for each validation sample and select, for metric computation, the sample that attains the highest VGG CS with respect to the ground-truth video.

PV output prediction. Reliability and sharpness are the two principal attributes of probabilistic forecasts. *Reliability* gauges the similarity between the predictive and observational distributions, whereas *sharpness* reflects the concentration of the predictive distribution. An informative probabilistic forecast must therefore be both reliable (encompassing the observations) and sufficiently sharp (narrow enough to convey information). In this study, the quality of photovoltaic (PV) output forecasts is measured by the *continuous ranked probability score* (CRPS) [49] and the *Winkler score* (WS) [50], both of which jointly assess reliability and sharpness.

The CRPS is defined as

$$\mathrm{CRPS} = \frac{1}{N}\sum_{k=1}^{N}\int_{-\infty}^{+\infty}\Big[F_{target}^k(x) - F_{model}^k(x)\Big]^2\,dx \tag{14}$$

where F_{target} is a Heaviside step function that jumps from 0 to 1 at the observed PV output, and F_{model} is the cumulative distribution function (CDF) of the model forecast. The CRPS, averaged over N prediction instances, shares the physical units of the target variable (kW in this work) and degenerates to the absolute error for deterministic forecasts, thereby enabling a direct comparison between probabilistic and point predictions [49].

The Winkler score at a nominal confidence level of $(1-\alpha)\%$ is expressed as

$$\mathrm{WS}_k = \begin{cases}\delta & \text{if } L_k \le x_k \le U_k\\ \delta + \dfrac{2(L_k - x_k)}{\alpha} & \text{if } x_k < L_k\\ \delta + \dfrac{2(x_k - U_k)}{\alpha} & \text{if } x_k > U_k\end{cases} \tag{15}$$

where $\delta = U_k - L_k$, and U_k and L_k denote the upper and lower bounds of the prediction interval, respectively. In this work, the 95th and 5th percentiles of the predictive distribution are adopted for U_k and L_k, resulting in a nominal 90 % confidence level ($\alpha = 0.1$). The score penalizes both intervals that fail to contain the observation x_k and unnecessarily wide intervals; hence, a lower WS implies a superior probabilistic forecast. The overall Winkler score across N samples is

$$\mathrm{WS} = \frac{1}{N}\sum_k^N \mathrm{WS}_k \tag{16}$$

To benchmark the proposed model against alternative approaches, we compute the *forecast skill* (FS), defined as

$$\mathrm{FS} = \left(1 - \frac{\mathrm{Error}_{forecast}}{\mathrm{Error}_{ref}}\right)\times 100\% \tag{17}$$

where the reference error corresponds to a baseline model, and the forecast error refers to the model under evaluation. In solar forecasting, the predominant baseline is the *smart persistence model* (SPM), which

assumes that the clear-sky index—the ratio of actual to theoretical clear-sky output—remains constant from time t to $t+T$:

$$k_{clr} = \frac{\mathcal{P}(t+T)}{\mathcal{P}_{clr}(t+T)} = \frac{\mathcal{P}(t)}{\mathcal{P}_{clr}(t)} \tag{18}$$

Here, k_{clr} is the clear-sky index, $\mathcal{P}$ is the measured PV output, and $\mathcal{P}_{clr}$ is the theoretical clear-sky output. The latter can be approximated for any timestamp using a clear-sky model based on solar geometry and panel orientation [51]:

$$P_{clr}(t) = P_m A_e \left\{ \cos\epsilon \cos\chi(t) + \sin\epsilon \sin\chi(t) \cos[\xi(t) - \zeta] \right\} \tag{19}$$

where $P_m = 1000\,\mathrm{W/m^{-2}}$ is the maximum solar irradiance, $A_e = 24.98\,\mathrm{m}^2$ is the effective panel area obtained via least-squares fitting to 12 clear-sky days [5], ϵ and ζ are the array tilt and azimuth angles (22.5° and 195°, respectively), and $\chi(t)$ and $\xi(t)$ are the solar zenith and azimuth angles, computable for any minute of the year from the empirical relations in [51]. Consequently, the T-minute-ahead SPM forecast is

$$\hat{\mathcal{P}}(t+T) = \frac{\mathcal{P}(t)}{\mathcal{P}_{clr}(t)} \times \mathcal{P}_{clr}(t+T) \tag{20}$$

Because SPM yields a single deterministic estimate, its CRPS reduces to the mean absolute error (MAE) [49]:

$$\mathrm{MAE} = \frac{1}{N} \sum_{i=1}^{N} |\hat{\mathcal{P}}_i - \mathcal{P}_i| \tag{21}$$

3. Experimental details and dataset

3.1. Experimental details

Diffusion model setup. This study uses a Diffusion Model for photovoltaic (PV) output forecasting. Following the approach of [52], we adopt a linear noise schedule and the noise prediction objective function described in Section 2.2. The number of diffusion steps T is set to 1000. To accelerate sampling, DPM-Solver++ [52] is employed as the sampling method.

To strengthen the justification of key design choices (diffusion steps/noise schedule, random-shift multimodal attention window sizes, and the mixed-mode loss weights in Eq. (12)), we conduct a dedicated sensitivity study under the main multimodal setting "SI PV→SI PV". Importantly, we avoid using the official validation split for ablation because that split is also used for reporting model performance and should remain unseen during any design selection. Instead, we construct an internal **ablation hold-out set** by selecting **two full days** from the training period and randomly sampling **250 time-consistent sequences per day** (500 samples in total), ensuring that each sampled instance satisfies the required chronological integrity (no missing frames/values within the context and prediction horizons). One day corresponds to **moderate** variability, and the other day represents a **high-difficulty** regime with stronger ramps and uncertainty. During the ablation stage, this hold-out set is **never used for optimization**; it is used only for comparing hyperparameter choices. For the **final model** reported on the official validation and test sets, this ablation hold-out set is merged back into the training set and used for training together with the rest of the training data. Unless otherwise specified, ablation comparisons use **10 stochastic forecast samples** per input to form an empirical predictive distribution, while the official validation/test evaluations follow Section 4.2 and generate **30 samples** per input for the three deployment configurations. Sampling is accelerated using **DPM-Solver++**. We select 60 solver steps based on a dedicated step-sensitivity experiment that jointly considers CRPS, **Winkler score (WS)**, and inference latency (see Section 4.3, Fig. 10 and Tables 7).

Model architecture. The overall forecasting framework consists of a coupled U-Net, which simultaneously predicts a future sky image sequence of shape $8\times3\times64\times64$ and a future PV sequence of shape 1×8, where the two sequences are temporally aligned. The coupled U-Net contains four scales, each comprising two standard PV-MM-Blocks and one down-/up-sampling block for the sky image sequence. At scales [2, 3, 4], *self-attention* and *cross-modal attention* are introduced for the sky image sequence; the window sizes for cross-modal attention are set to [1, 4, 8]. The choice of these window sizes is supported by the sensitivity analysis reported in Section 4.3.

Table 1
Measured training/inference time of PV-MM-Diffusion (wall-clock).

Stage	Hardware	Wall-clock time
Training (final model, 100,000 steps)	8× NVIDIA RTX 3090	43.06 h
Inference (60 solver steps, 10 samples)	1× NVIDIA RTX 3090	45.3895 s/instance

Table 2
Accuracy–latency trade-off by varying the number of stochastic samples (test set; DPM-Solver++ with 60 solver steps).

# Samples N_{samp}	CRPS [kW] ↓	WS [kW] ↓	Relative latency†
30 (benchmark setting)	2.63	21.46	≈ 3×
10 (deployment-friendly)	2.78	25.65	1×
5 (ultra-fast)	3.01	34.56	≈ 0.5×

† Relative latency is normalized to the 10-sample setting; in practice, latency grows approximately linearly with N_{samp} under the same solver-step budget and can be reduced via multi-GPU parallel sampling.

3.2. Computational complexity and practical deployment

Given that diffusion models generate probabilistic forecasts through iterative denoising, the computational cost is dominated by repeated forward passes of the Coupled U-Net during sampling. With DPM-Solver++, the inference cost scales approximately as $\mathcal{O}(N_{\mathrm{samp}} \cdot N_{\mathrm{solver}})$, where N_{samp} is the number of stochastic samples used to form an empirical predictive distribution and N_{solver} is the number of solver steps. In addition, the memory footprint is mainly determined by the Coupled U-Net activations and therefore scales with the image resolution and the multi-scale feature channels.

Measured wall-clock time. Training the final PV-MM-Diffusion model for 100,000 optimization steps takes **43.06 hours** on **8× NVIDIA RTX 3090 GPUs**. For inference, using **DPM-Solver++ with 60 solver steps**, generating a predictive distribution with **10 samples** takes **45.3895 s** per forecast instance on **1× NVIDIA RTX 3090** (averaged over instances).

Suitability for real-time operation. Our task predicts the next 15 minutes and can be updated at the data cadence; therefore, the above latency is already within the operational horizon. For stricter real-time constraints, the sampling budget can be reduced (fewer samples and/or fewer solver steps) with a controlled loss in probabilistic accuracy, and sampling is *embarrassingly parallel* across samples. Thus, practitioners can further reduce latency by (i) allocating multiple GPUs to generate different samples in parallel, and/or (ii) batching multiple PV sites on the same GPU server.

Accuracy–latency trade-off. Table 2 reports the test-set performance under different sampling budgets (solver steps fixed to 60). Importantly, even the fast settings remain competitive: with 10 samples we obtain CRPS = 2.78 kW and WS = 25.65 kW, and with 5 samples CRPS = 3.01 kW and WS = 34.56 kW, both outperforming the SkyGPT→U-Net baseline (CRPS = 3.21 kW, WS = 35.62 kW).

3.3. Sky image and PV dataset

This study directly uses the preprocessed and partitioned sky images and PV power dataset provided by SkyGPT [15]. The sky images were collected by a fisheye camera installed on the roof of the Green Earth Sciences Building at Stanford University, covering the period from

March 2017 to December 2019. The PV power data are from a 30 kW rooftop PV system adjacent to the camera (the original data source can be found in [53]). After downsampling the original 2048×2048 images to 64×64, researchers aligned the timestamps of the sky images and PV power at a frequency of one frame per minute to reduce computational overhead.

To focus on multi-cloud conditions and ensure continuous time series information, the researchers employed a modified normalized red-blue ratio method [54] to exclude clear-sky samples, retaining only images and PV power data under multi-cloud conditions. Subsequently, to ensure the completeness of the time series in both the previous and subsequent 15 minutes, the filtered data were sampled at 2-minute intervals, yielding 60,385 valid records. Each record contains the past and future 15-minute sky image sequences and corresponding PV sequences.

On this basis, the dataset is divided into training, validation, and test sets. The training set contains 53,336 samples (approximately 88 %), the validation set 4467 samples (approximately 7 %), and the test set 2582 samples (approximately 4 %). The validation set is selected from 10 multi-cloud days between March 2017 and October 2019, considering both seasonal and annual variations. The test set comes from 5 multi-cloud days in November and December 2019 to evaluate the model's generalization capability. Neither the validation set nor the test set is used for model training.

This paper fully adopts the aforementioned data partitioning and preprocessing results, without any additional filtering or segmentation. For more details on the dataset, refer to [15].

Scope and implications for generalization. We emphasize that the above dataset corresponds to a *single* site (Stanford University, California, USA), with one ground-based fisheye sky camera and one adjacent 30 kW rooftop PV system. Although the training/validation/test split spans multiple seasons and years, it does not capture cross-site variability in cloud regimes (e.g., tropical convection or monsoon-driven cloud fields), terrain-induced horizon shading, or PV-system characteristics (e.g., tracking systems, inverter clipping/curtailment, and different module technologies). Therefore, the reported results mainly demonstrate *within-site* temporal generalization rather than *cross-site* generalization. The expected domain-shift issues and practical adaptation strategies are explicitly discussed in Section 5.

4. Results

4.1. Sky video prediction

We evaluate sky-image sequence forecasting from two complementary perspectives: *accuracy* (closeness to the ground truth) and *diversity* (coverage of multiple plausible cloud-evolution scenarios). Figs. 4–6 summarize the evidence: Fig. 4 provides qualitative comparisons on representative cloud-transition cases; Fig. 5 reports VGG cosine similarity (VGG CS) trends over lead times for both accuracy and diversity; and Fig. 6 visualizes multiple sampled futures under the same historical context. For stochastic generators, we use multiple samples per input (SkyGPT: 10; PV-MM-Diffusion: 30) and either select the best-matching sample for accuracy reporting or aggregate all samples to quantify diversity.

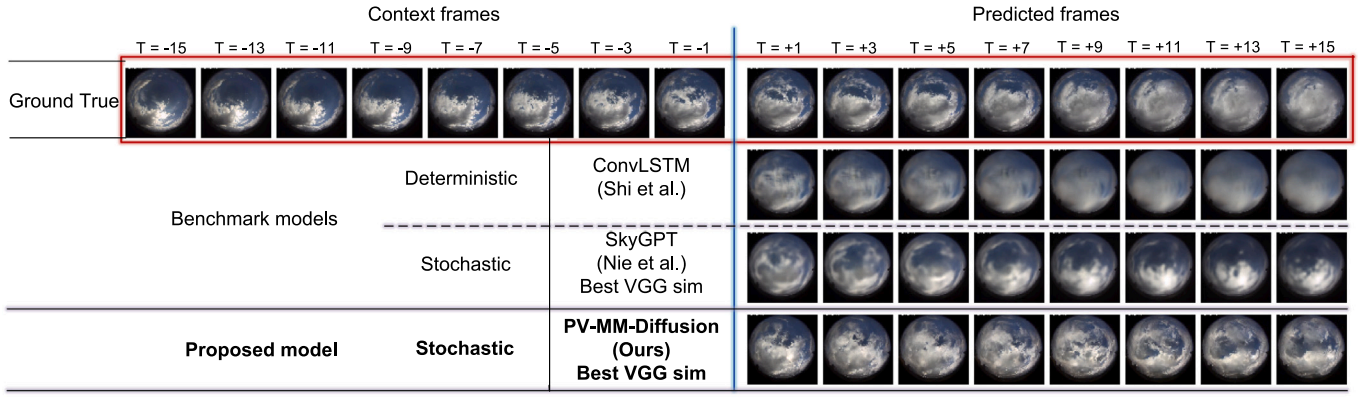
Overall, ConvLSTM tends to lose fine-grained textures and becomes increasingly blurred at longer horizons, while SkyGPT better preserves large-scale advection but can miss small-scale illumination and cloud-edge details. PV-MM-Diffusion produces sharper cloud boundaries and more realistic lighting cues throughout the 15-minute horizon, consistent with its higher VGG CS in Fig. 5(a). In terms of diversity, the dispersion in Fig. 5(b) increases with lead time for both generative models, and PV-MM-Diffusion exhibits comparable or larger variability, indicating richer multimodal futures. The case study in Fig. 6 further illustrates that PV-MM-Diffusion can generate distinct yet physically plausible trajectories while maintaining the dominant cloud-motion trend observed in the ground truth.

4.2. Probabilistic solar forecasting

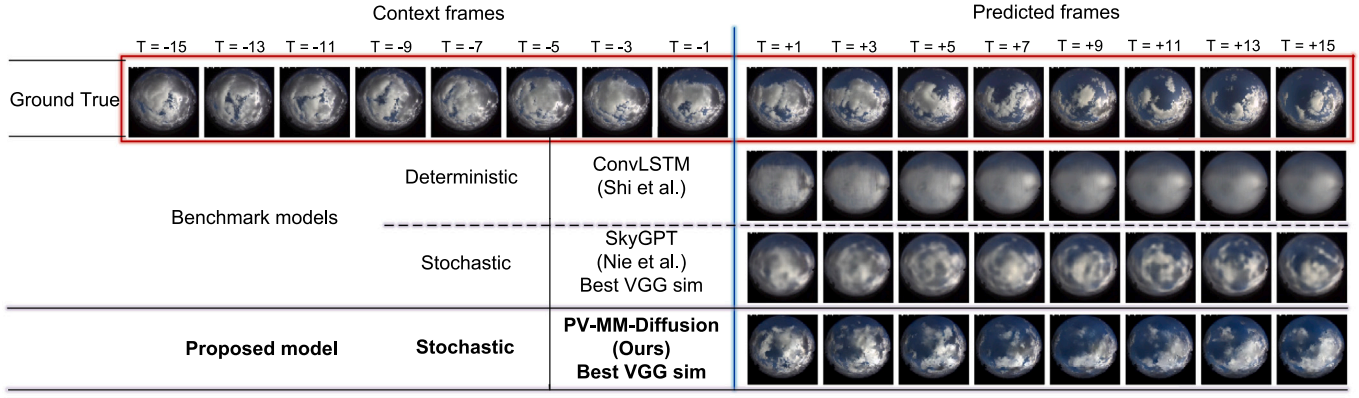
In this study, we focus on a 15-minute-ahead photovoltaic (PV) forecasting task. Our target system is a 30 kW rooftop PV installation on a university campus (see Section 3.3 for data details). Given the increasing penetration of renewable energy in power grids, prediction intervals offer greater risk management value than single-point predictions. Hence, this study adopts interval forecasting: by repeatedly sampling, multiple point forecasts are generated and subsequently used to construct the final prediction intervals. We note that the experimental validation in this work relies on a single rooftop PV system at one site; therefore, generalization to other climates, PV technologies, or larger spatial scales is not guaranteed without additional multi-site data and re-training/fine-tuning. We explicitly acknowledge this limitation and discuss expected performance under alternative sensing setups (e.g., satellite imagery or PV-only operation) in Section 5.

To comprehensively evaluate the performance of the proposed method under different scenarios, we select the following representative benchmark models, and we additionally include two widely used probabilistic baselines (MC Dropout and quantile regression) to strengthen the benchmark set:

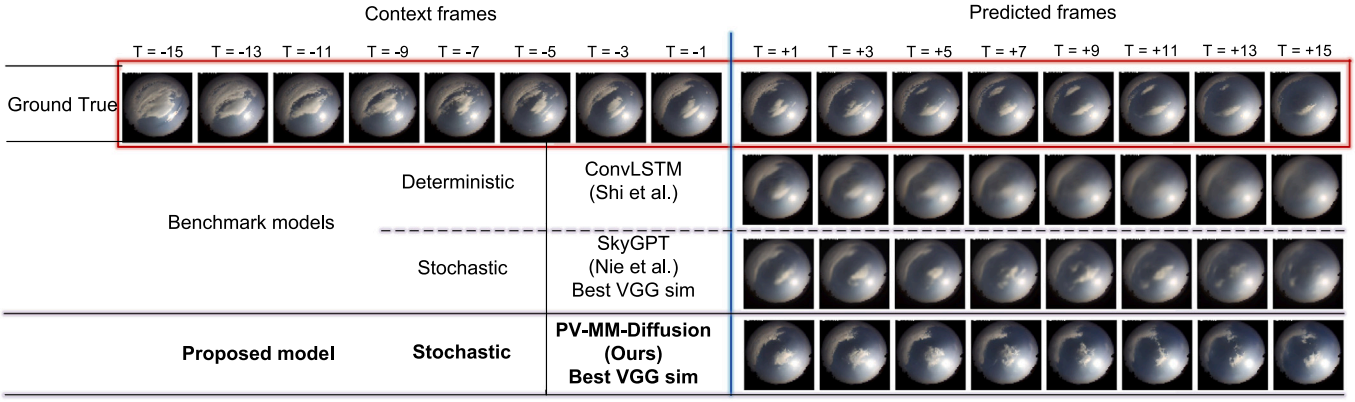
1. **Smart Persistence Model (SPM).** The specific definition is given in Eqs. (18)–(20). This model assumes that the relative power output or clear-sky index (CSI, defined as the ratio of the actual power output to the theoretical clear-sky power output) remains constant over the 15-minute prediction horizon (i.e., $CSI_t = CSI_{t+15}$). Hence, it can only generate a single-point forecast at a given timestamp. Despite its simplicity and frequent use as a baseline, SPM neglects the stochastic evolution of cloud cover and thus struggles to capture large power fluctuations during partly cloudy conditions.
2. **Conventional end-to-end baseline model SUNSET** [5]. This model employs a convolutional neural network (CNN) architecture similar to AlexNet [55], taking as input a sequence of sky images and PV power outputs from the past 15 minutes (from $t-15$ to t, with a temporal resolution of 1 minute), and directly outputting the PV power at the future time step $t+15$. Using 10-fold cross-validation yields 10 sub-models, each making a prediction for the same input, resulting in 10 PV output values. Although SUNSET outperforms SPM in capturing short-term trends, it is still inadequate in depicting uncertainty under highly dynamic, partially cloudy conditions.
3. **MC Dropout SUNSET.** To obtain a stronger uncertainty-aware variant of SUNSET, we adopt Monte Carlo (MC) dropout at inference time, which interprets dropout as approximate Bayesian inference and enables predictive sampling without changing the original network structure [56]. Concretely, we keep SUNSET's training protocol and draw multiple stochastic forward passes with dropout enabled to form an empirical predictive distribution (30 samples in our evaluation).
4. **QR-GRU (PV-only quantile regression baseline).** We further implement a quantile-regression recurrent baseline that takes only historical PV measurements as input and directly outputs multiple predictive quantiles. Specifically, a two-layer GRU [57] is trained with the pinball loss to estimate the conditional quantiles [58]. Following prior PV uncertainty-quantification practice [28], we compute CRPS and Winkler score from the resulting predictive quantiles by sampling 30 values from the implied distribution, ensuring a fair metric protocol consistent with other stochastic methods.
5. **Three two-stage "sky image prediction + PV mapping" models.** This type of model first forecasts or generates future sky images, and then inputs these predicted images into a PV power forecasting module. Specifically, *SkyGPT→U-Net* [15] employs the generative video prediction model SkyGPT in the first stage;



（a）The sky condition changing from partly cloudy to overcast



（b）The sky condition changing from overcast to partly cloudy



（c）The sky condition changing from partly cloudy to clear sky

Fig. 4. Representative qualitative comparison of future sky-image prediction on three validation samples with different cloud regimes: (a) partly cloudy → overcast, (b) overcast → partly cloudy, and (c) partly cloudy → clear within the next 15 min. Left: context frames ($t-16$ to t min); right: future frames ($t+1$ to $t+15$ min). The first row is the ground-truth sequence, and the remaining rows are forecasts from different models. For stochastic models (SkyGPT and PV-MM-Diffusion), multiple future sequences are sampled (10 and 30 samples, respectively) and the sample with the highest VGG cosine similarity (VGG CS) to the ground-truth future sequence is reported ("Best VGG sim.").

for the same historical sky image sequence, 10 random samples are drawn to generate 10 possible future sky image sequences. Each sampled future image at time $t + 15$ is then fed into the improved U-Net based PV prediction model. Since U-Net adopts 10-fold cross-validation, there are 10 sub-models, and combining these with the 10 future scenarios yields a total of 100 predictions.*ConvLSTM→U-Net* uses a deterministic video prediction model (ConvLSTM) to produce a single future sky image sequence, which is then fed into the same U-Net sub-models (also obtained via 10-fold cross-validation), resulting in 10 PV predictions. Because ConvLSTM lacks generative sampling capabilities, it is limited in capturing the diversity of future cloud conditions. To assess the upper bound of a two-stage framework under ideal conditions, we set up *Real future SI→U-Net*, which directly uses the real sky image at $t + 15$ instead of performing any video prediction. Although this is not feasible in practical applications, it represents the potential best case for two-stage methods if the video prediction component were error-free. *Real future SI→U-Net* also uses 10-fold cross-validation to yield 10 PV forecasts.

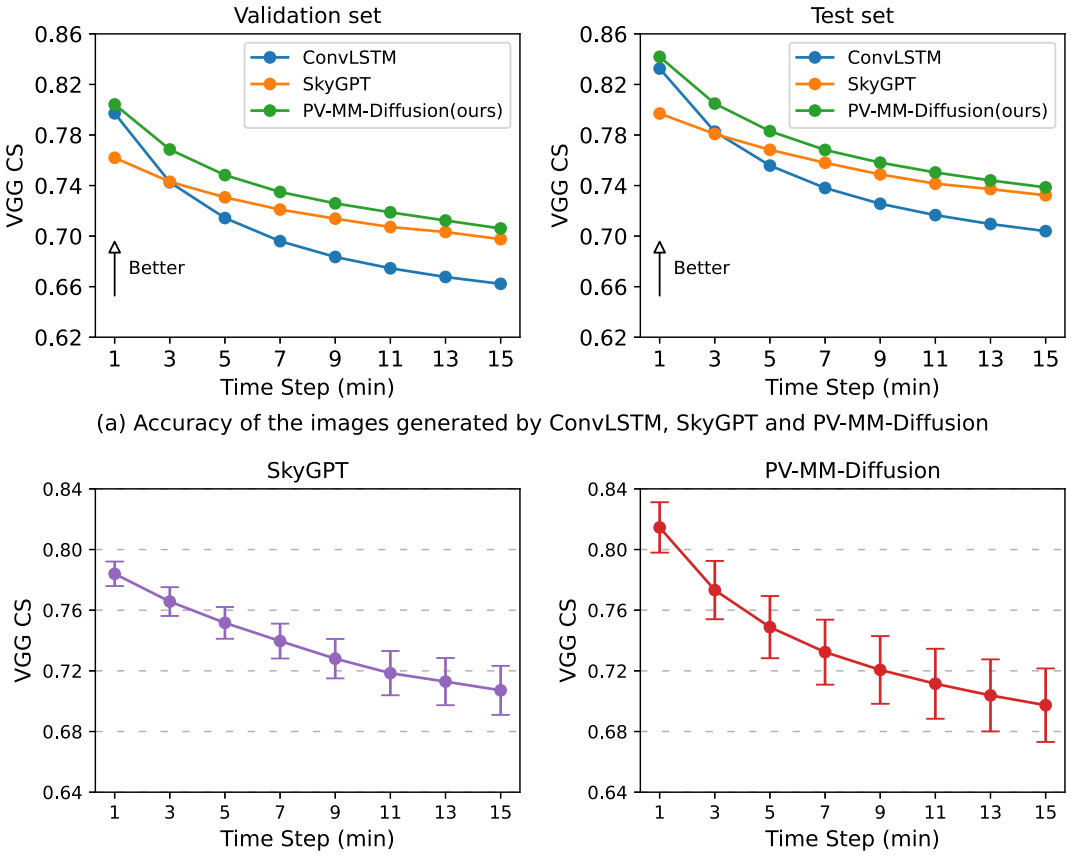


Fig. 5. Quantitative evaluation of images generated by PV-MM-Diffusion and benchmark video prediction models at different time steps. **(a) Accuracy Evaluation:** For the generative models SkyGPT and PV-MM-Diffusion, 10 and 30 future prediction sequences were generated for each sample in the validation and test sets, respectively. The sequence that is closest to the real images was then selected to compute the average accuracy. **(b) Diversity Evaluation:** Conducted only on the test set, for each sample, the predicted sequences were used to calculate the average VGG CS (data points) and the corresponding standard deviation (error bars). It can be observed that as the prediction steps extend from the near future to the more distant future, the error bars gradually increase, indicating that the model's predictions exhibit higher variability at later time steps.

Building on these comparison methods, we further test the proposed framework under different input and output configurations, as introduced in Section 2.5: *PV-MM-Diffusion* ("SI PV → SI PV"), "SI → SI PV" and "PV → PV". For the main benchmark comparison with prior work, we generate **30** stochastic samples per input to form the empirical predictive distribution (Table 3). For practical deployment, the sampling budget can be reduced (e.g., **10** or **5** samples) to meet latency constraints, and we report the corresponding accuracy–latency trade-off in Section 3.2 (Table 2). "SI PV → SI PV" takes both historical sky images and historical PV sequence as input, and outputs both future sky images and PV sequence; "SI → SI PV" takes only the historical sky image sequence as input and outputs both future sky images and future PV sequence; and "PV → PV" takes only the historical PV sequence as input and outputs only the future PV sequence.

We evaluate all methods on the official validation set (4,467 samples from 10 cloudy days) and test set (2,582 samples from 5 cloudy days) following the SkyGPT split (Section 3.3). Probabilistic performance is assessed using CRPS and Winkler score (WS) as defined in Section 2.6. Forecast skill (FS) is computed relative to SkyGPT→U-Net using Eq. (17).

Table 3 presents the CRPS, FS (CRPS), WS, and FS (WS) results for each model on the validation and test sets. It can be seen that the "SI PV → SI PV" configuration delivers the best performance. Compared to the two-stage methods, it exhibits significant improvements in all metrics. It might be because two-stage methods typically generate future sky images first, and then feed these images into a PV forecasting model; the video prediction step does not simultaneously consider PV power output, resulting in a fragmented learning process between future sky image generation and PV forecasting. In contrast, "SI PV → SI PV" simultaneously uses historical PV and sky image sequences as conditions to jointly generate future PV and sky image sequences in a latent space, effectively avoiding the information fragmentation found in two-stage models.

From Fig. 7, one can observe that if the real sky image at time $t+15$ is directly fed into U-Net (*i.e.*, the "ideal" two-stage method), CRPS and WS are improved by about 21 % and 14 %, respectively, compared to *SkyGPT→U-Net*. This indicates that there is still substantial room for improvement in the video prediction component (carried out by *SkyGPT*). When comparing the "ideal" two-stage model with "SI PV → SI PV," the former's CRPS is only about 3 % higher, yet its WS is 26 % worse, suggesting that the multimodal diffusion model has stronger overall performance. Notably, even when the two-stage framework uses the real sky image at $t+15$, the CRPS and WS still show respective potential improvements of 79 % and 86 %, implying that the mapping step from sky image to PV power remains a major source of error. Therefore, even if the video prediction stage is assumed to be "perfect," the subsequent power mapping could still introduce bias. In contrast, "SI PV → SI PV" explicitly leverages the prior information from historical PV sequences during forecasting, significantly reducing overall errors. Its CRPS is already close to that of the "ideal" two-stage method, while its WS is markedly better. This demonstrates that the multimodal information exchange between sky image sequences and PV sequences can be complementary, thereby substantially enhancing forecasting accuracy for PV power.

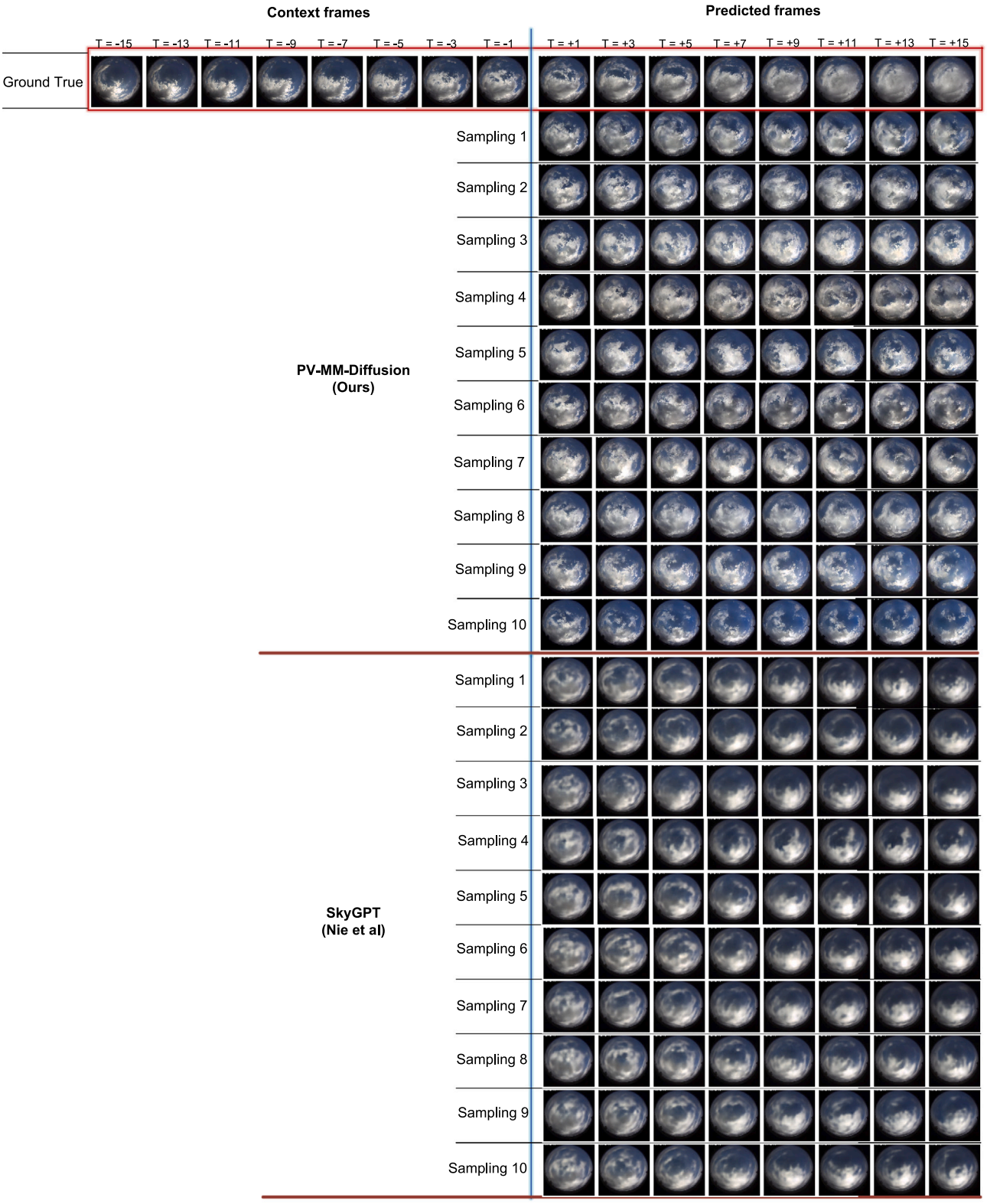


Fig. 6. Diversity of generated future sky-image scenarios under identical context frames (validation example). For each model, 10 stochastic samples are drawn. PV-MM-Diffusion produces a wider range of plausible cloud-motion and illumination patterns at mid-to-long horizons while remaining consistent with the Given context.

Fig. 8 illustrates the forecast curves generated by different methods on three partly cloudy days in the test set. PV output tends to fluctuate more sharply under partly cloudy conditions. To intuitively show the uncertainty in the forecasts, two confidence intervals are plotted: the light blue shading represents the 5–95 percentile range (i.e., the 90 % confidence interval), and the dark blue shading represents the 25–75 percentile range (i.e., the 50 % confidence interval). From top to bottom, the first to sixth rows show the probabilistic baselines and two-stage

Table 3
This table summarizes the performance of various solar forecasting methods on the validation and test sets using CRPS, FS(CRPS), WS, and FS(WS) as evaluation metrics. As baseline references, this study includes Smart Persistence, SUNSET, MC Dropout SUNSET, QR-GRU, ConvLSTM(D) → U-Net, SkyGPT(S) → U-Net, and a two-stage model using real future sky images (Real future SI → U-Net), where (D) and (S) represent deterministic and stochastic predictions, respectively. The three proposed configurations in this study are: "SI PV → SI PV (S) (Ours)", "SI → SI PV (S) (Ours)", and "PV → PV (S) (Ours)". The best-performing models in each metric are in bold. For the "ideal" two-stage model based on real future sky images, pink and green cells denote performance that is worse and better, respectively, compared to that approach.

Method	Validation Set		Test Set	
	CRPS [kW] ↓ (FS [%] ↑)	WS [kW] ↓ (FS [%] ↑)	CRPS [kW] ↓ (FS [%] ↑)	WS [kW] ↓ (FS [%] ↑)
Smart Persistence	2.89 (−21 %)	–	3.67 (−14 %)	–
SUNSET	2.50 (−5 %)	42.33 (−73 %)	3.28 (−2 %)	56.29 (−58 %)
MC Dropout SUNSET	2.31 (4 %)	29.72 (−22 %)	3.10 (3 %)	43.68 (−23 %)
QR-GRU	2.21 (7 %)	22.32 (9 %)	2.95 (8 %)	33.38 (6 %)
ConvLSTM (D) → U-Net	2.91 (−22 %)	45.16 (−85 %)	3.12 (3 %)	46.33 (−30 %)
SkyGPT (S) → U-Net	2.39 (0 %)	24.44 (0 %)	3.21 (0 %)	35.62 (0 %)
Real future SI → U-Net	1.71 (28 %)	20.47 (16 %)	2.52 (21 %)	30.72 (14 %)
"SI PV → SI PV (S) (Ours)"	**2.09 (13 %)**	**16.50 (32 %)**	**2.63 (18 %)**	**21.46 (40 %)**
"SI → SI PV (S) (Ours)"	2.33 (3 %)	18.35 (25 %)	3.09 (4 %)	24.99 (30 %)
"PV → PV (S) (Ours)"	2.26 (5 %)	23.96 (2 %)	3.02 (6 %)	35.86 (−1 %)

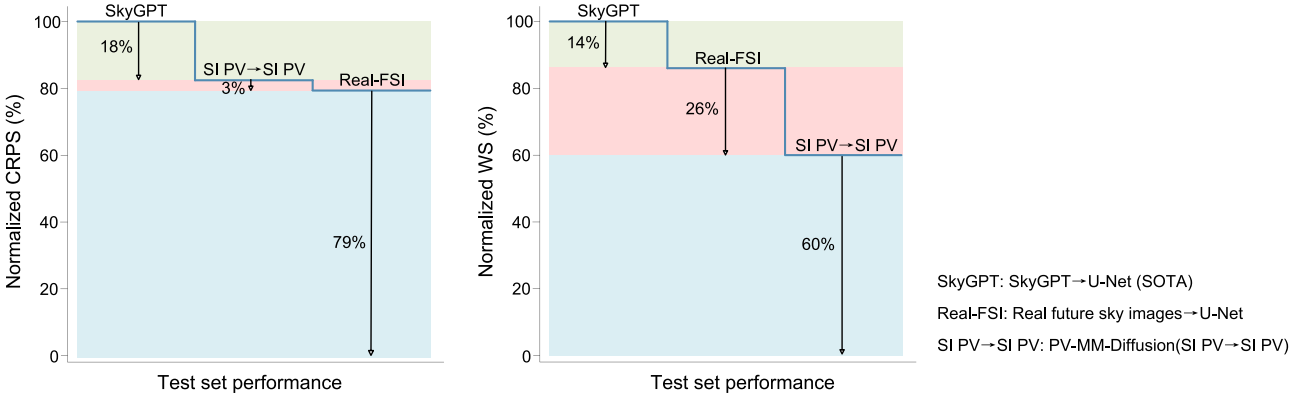


Fig. 7. Comparative performance on the test set of the proposed *PV-MM-Diffusion* approach ("*SI PV → SI PV*") and the two-stage baselines (*SkyGPT → U-Net* and *Real future sky images → U-Net*). To facilitate an intuitive comparison, the continuous ranked probability score (CRPS) and the Winkler score (WS) shown in the Figure are normalized with respect to the *SkyGPT* baseline (i.e., CRPS/$\text{CRPS}_{\text{SkyGPT}}$ and WS/$\text{WS}_{\text{SkyGPT}}$). The relative improvements are highlighted by staircase plots.

baselines: *SUNSET*, *MC Dropout SUNSET*, *QR-GRU*, *ConvLSTM→U-Net*, *SkyGPT→U-Net*, and *Real future SI→U-Net*, respectively. The remaining rows report the proposed diffusion configurations ("*SI PV → SI PV*"; "*SI → SI PV*"; and "*PV → PV*").

It can be observed that *SUNSET* performs poorly in capturing ramp events, particularly on days such as 2019-12-23 where PV output exhibits large fluctuations. In these cases, the model's prediction of peak and abrupt drop events lags considerably behind the actual measurements, and the overall predictive intervals are quite narrow. This indicates that the diversity among the 10 sub-models in the ensemble is limited. Compared with SUNSET, QR-GRU yields more competitive uncertainty bounds, while MC Dropout SUNSET provides an additional deep-learning probabilistic baseline via stochastic dropout sampling. By contrast, the two-stage methods (both deterministic and generative) exhibit broader predictive intervals overall, suggesting that their sub-model ensembles are more diverse than those in *SUNSET*. Nevertheless, even the "ideal" two-stage method (*Real future SI→U-Net*) performs unsatisfactorily on days with large PV output variability: ramp events are neither well captured by the median forecast nor covered by the predictive intervals. In contrast, the end-to-end multimodal diffusion model ("*SI PV → SI PV*") is more effective at capturing ramp events: its median forecast is typically located around the middle of the fluctuation range, the prediction interval narrows when PV output fluctuations are small, and it widens notably when fluctuations are large. This behavior aligns with the expected dynamic changes in uncertainty. By comparison, the single-modality configurations ("*SI → SI PV*" and "*PV → PV*") cannot fully exploit the potential prior knowledge contained in the other modality, making them less capable of capturing ramp events. They often need to expand their predictive intervals to reduce the risk of missing large deviations. Overall, multimodal fusion ("*SI PV → SI PV*") outperforms the single-modality configurations in terms of both the accuracy of amplitude and timing of the fluctuations, as well as the completeness of interval characterization.

Fig. 9 displays how the CRPS and WS metrics change as the number of future samples increases for different configurations: the dual-modality "SI PV → SI PV", and the single-modality "SI → SI PV" and "PV → PV". It can be seen that for all configurations, both CRPS and WS consistently decrease as the number of samples grows; however, once the sample count exceeds a certain threshold, the performance gains begin to diminish, indicating a trade-off between predictive accuracy and computational overhead. Notably, under the same number of samples, the dual-modality configuration ("*SI PV → SI PV*") always achieves lower CRPS and WS values, showing a clear advantage over the single-modality methods.

4.3. Ablation and sensitivity analysis for key design choices

This section provides a targeted sensitivity analysis to justify key design choices, including: (i) the mixed-mode loss weights among the three

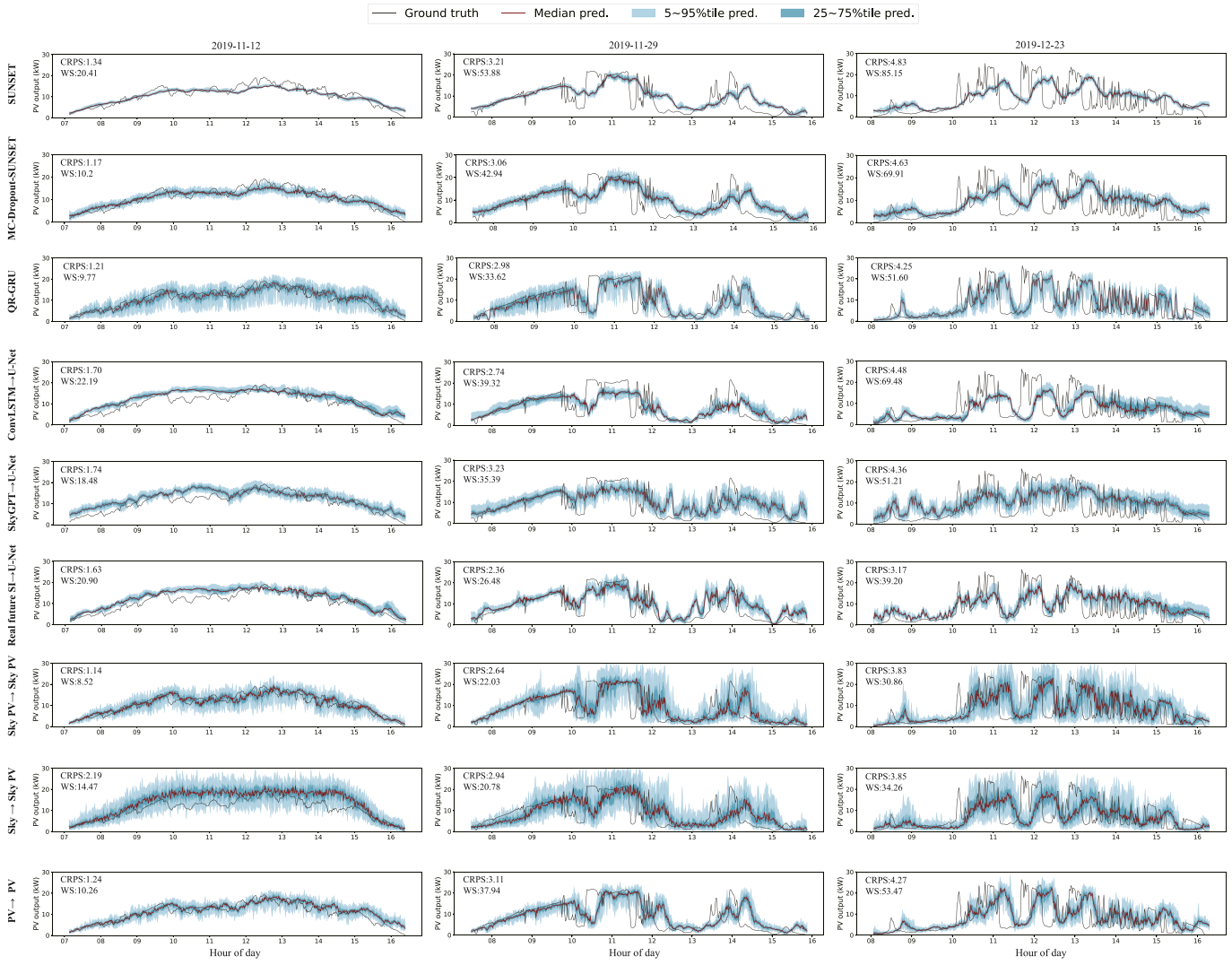


Fig. 8. Probabilistic PV power forecasts on three representative partly cloudy test days. All curves are aligned to the *target timestamps* (i.e., the prediction at horizon t+15 is plotted at time t+15 on the x-axis). Black line: measured PV power; red line: median forecasts. Shaded bands denote central prediction intervals computed from 30 stochastic samples per input: dark blue band = 50 % interval (25th–75th percentiles) and light blue band = 90 % interval (5th–95th percentiles). Rows correspond to SUNSET, MC-Dropout-SUNSET, QR-GRU, ConvLSTM→U-Net, SkyGPT→U-Net, real future SI→U-Net, and the three PV-MM-Diffusion deployment modes (Sky PV→Sky PV, Sky→Sky PV, PV→PV).

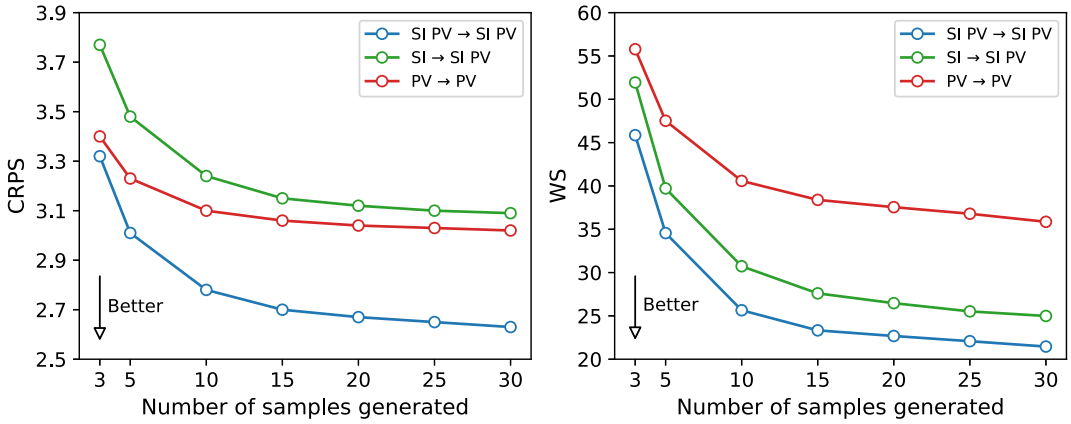


Fig. 9. Effect of the number of stochastic samples on probabilistic PV forecasting performance (test set). CRPS and Winkler score (WS) are computed from the empirical predictive distribution formed by N_{samp} reverse-diffusion samples for each deployment mode. Increasing N_{samp} improves both metrics with diminishing returns, illustrating the accuracy–latency trade-off.

training modes in Eq. (12), (ii) the number of diffusion steps and the noise schedule, and (iii) the window sizes used in the random-shift multimodal attention (RS-PV-MMA). All experiments are conducted under the **SI PV → SI PV** configuration and evaluated on the internal **ablation hold-out set** described in Section 3.1. Unless otherwise stated, probabilistic metrics are computed from **10 stochastic samples** per input. Inference uses **DPM-Solver++ with 60 solver steps** for all settings to ensure comparability. When a setting is obtained via continued training rather than training from scratch, we explicitly state it in the corresponding subsection.

4.3.1. Sensitivity to mixed-mode loss weights

We study how the mixture weights $\boldsymbol{w} = (w_{\mathrm{SP}}, w_{\mathrm{SI}}, w_{\mathrm{PV}})$ in Eq. (12) affect probabilistic forecasting quality and robustness across the three deployment modes. The weights are reported in the fixed order **SI PV→SI PV/SI→SI PV/PV→PV**. The default setting 0.33_0.33_0.33 follows the original design and is trained for 100,000 steps. The re-weighted variants 0.50_0.25_0.25, 0.25_0.50_0.25, and 0.25_0.25_0.50 are obtained by **continuing training for 20,000 additional steps** starting from the default checkpoint, after switching $\boldsymbol{w}$. We also include 1_0_0 as a single-mode baseline trained **from scratch** for 100,000 steps (only optimizing the SI PV→SI PV term). All settings are evaluated on the ablation hold-out set (500 samples) without any optimization on that set. At inference, we draw 10 stochastic samples per input to form an empirical predictive distribution, using DPM-Solver++ (60 solver steps).

Table 4 indicates that changing $\boldsymbol{w}$ can yield gains for a specific metric in a specific mode (e.g., emphasizing the PV→PV term improves WS under SI PV→SI PV), but often degrades performance under other deployment modes, especially those requiring missing-modality robustness (SI→SI PV and PV→PV). Notably, even with an additional 20,000 fine-tuning steps, none of the re-weighted variants consistently outperforms the balanced setting across all modes. Therefore, we retain 0.33_0.33_0.33 as the default choice in the main experiments for robust multi-mode deployment.

4.3.2. Diffusion steps and noise schedule

We first study the sensitivity to the diffusion configuration by varying (a) the total number of diffusion steps T and (b) the noise schedule. Table 5 reports the results. Reducing the diffusion steps from $T = 1000$ to $T = 500$ under a linear schedule degrades both CRPS and Winkler score, suggesting that an overly short diffusion trajectory provides insufficient granularity for stable denoising and uncertainty calibration. When keeping $T = 1000$ but switching from a linear to a cosine schedule, CRPS slightly improves, but the Winkler score becomes substantially worse, indicating a less favorable reliability–sharpness trade-off of the resulting prediction intervals. Overall, $T = 1000$ **with a linear noise schedule** yields the best interval quality (lowest Winkler score) while maintaining competitive CRPS, and we therefore adopt this configuration in all main experiments.

Table 4
Ablation on mixed-mode loss weights $\boldsymbol{w}$ in Eq. (12) (ablation hold-out set; 10 samples per input; DPM-Solver++ single-step with 60 solver steps). Lower is better.

Weights (SP/SI/PV)	SI PV→SI PV		SI→SI PV		PV→PV	
	CRPS ↓	WS ↓	CRPS ↓	WS ↓	CRPS ↓	WS ↓
0.33_0.33_0.33 (default)	1.92	18.02	2.33	**18.76**	**2.19**	24.01
0.50_0.25_0.25	**1.91**	17.91	2.65	23.42	2.20	26.55
0.25_0.50_0.25	1.94	18.04	**2.30**	19.31	2.22	**23.82**
0.25_0.25_0.50	1.97	**15.40**	3.36	30.01	2.25	27.58
1_0_0	2.06	18.90	–	–	–	–

Table 5
Ablation on diffusion steps and noise schedule (SI PV → SI PV, 10 samples per input, DPM-Solver++ 60 steps).

Setting	Steps T	Schedule	Train Steps	CRPS [kW] ↓	WS [kW] ↓
T500_linear	500	Linear	40,000	2.41	25.03
T1000_linear (default)	1000	Linear	40,000	2.38	**23.73**
T1000_cosine	1000	Cosine	40,000	**2.33**	30.24

Table 6
Ablation on RS-PV-MMA window sizes (SI PV → SI PV, 10 samples per input, DPM-Solver++ 60 steps).

Window sizes (top/mid/bottom scales)	Train Steps	CRPS [kW] ↓	WS [kW] ↓
[1, 2, 4]	100,000	1.93	19.11
[1, 4, 8] (default)	100,000	**1.92**	**18.02**
[2, 4, 8]	100,000	1.94	18.16

4.3.3. Window sizes in random-shift multimodal attention (RS-PV-MMA)

We further evaluate the sensitivity of RS-PV-MMA to the cross-modal attention window sizes used at different U-Net scales. We compare three window-size triplets: **[1,2,4]** , **[1,4,8]** (our default), and **[2,4,8]** , while keeping all other settings unchanged. The results are summarized in Table 6. Among the tested options, **[1,4,8]** achieves the best CRPS and the best Winkler score. Using smaller windows ([1,2,4]) slightly worsens both metrics, likely because deeper layers cannot aggregate sufficiently long-range cross-modal context. Conversely, increasing the smallest (finest-scale) window from 1 to 2 ([2,4,8]) also degrades performance, which is consistent with the intuition that the upper layers benefit from very local alignment, while larger windows are more suitable for deeper layers that capture higher-level semantics. Based on this sensitivity study, we keep **[1,4,8]** as the default window-size configuration in RS-PV-MMA.

Note that the absolute values in Tables 5 and 6 are reported on an internal ablation hold-out set drawn from the training period and are therefore not directly comparable to the official validation/test results in Table 1; our goal here is to quantify the *relative* sensitivity under a strict no-peeking protocol.

4.3.4. Number of sampling steps in DPM-solver++

We further investigate the effect of the *sampling steps* (i.e., the number of solver steps in DPM-Solver++) on probabilistic forecasting quality and inference latency under the main multimodal setting **SI PV→SI PV**. In this study, we keep the inference protocol fixed: **10 stochastic samples** are drawn for each input to form an empirical predictive distribution.

As shown in Fig. 10 and Table 7, increasing the sampling steps generally improves probabilistic accuracy (lower CRPS and WS) but leads to a near-linear increase in inference time. While 70 steps achieve the lowest CRPS, its WS is slightly worse than 60 steps and the latency is noticeably higher. Therefore, we adopt **60 sampling steps** as the default setting, which yields the best overall trade-off between distributional fidelity (especially WS) and computational cost.

5. Limitations and future work

In this study, a multi-modal photovoltaic forecasting framework based on diffusion models (PV-MM-Diffusion) was constructed, achieving certain progress in ultra-short-term probabilistic forecasting. Meanwhile, it also reveals the following limitations and directions for improvement:

1. **Limited dataset scope and generalization across climates, PV technologies, and spatial scales:**
   The current experimental validation is conducted on *one* rooftop PV system co-located with *one* sky camera at a single site. Hence, while the test set is temporally out-of-sample, the reported results primarily demonstrate *within-site* generalization across

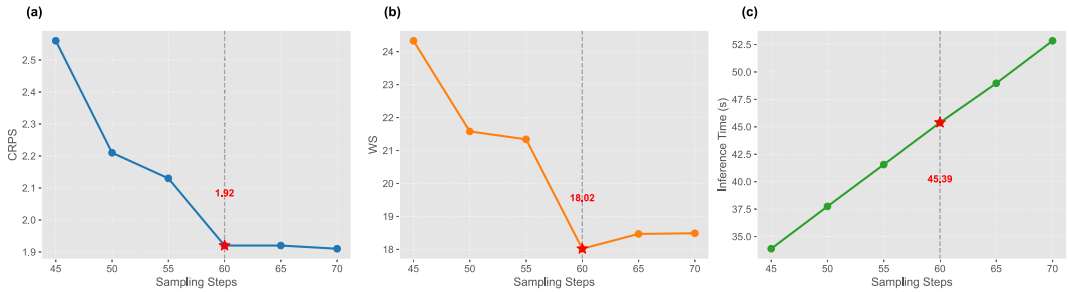


Fig. 10. Ablation study on the number of sampling steps (DPM-Solver++ solver steps). (a) CRPS trend; (b) Winkler score (WS) trend; (c) inference time for generating 10 samples. The red star indicates the chosen configuration (60 steps), which provides the best trade-off between probabilistic performance and computational efficiency.

Table 7
Quantitative results for different sampling steps (SI PV→SI PV; 10 samples per input).

Sampling steps	CRPS [kW] ↓	WS [kW] ↓	Inference time (10 samples) [s] ↓
45	2.56	24.33	33.89
50	2.21	21.58	37.75
55	2.13	21.34	41.56
60	1.92	**18.02**	45.39
65	1.92	18.47	48.97
70	**1.91**	18.49	52.84

seasons/years, rather than *cross-site* generalization. In practice, transferring the framework to new regions may face domain shifts in (i) cloud dynamics and illumination conditions (e.g., coastal stratocumulus vs. inland convection, aerosol conditions, and seasonal cloud regimes), (ii) PV-system characteristics (e.g., module technology, tilt/azimuth, tracking, temperature sensitivity, inverter clipping/curtailment, and partial shading), and (iii) scale effects (e.g., spatial smoothing in utility-scale plants or feeder/region-level aggregation). These factors can change the mapping between observed clouds and PV output and may affect both accuracy and uncertainty calibration.

This study used 53,336 samples under cloudy conditions to train the PV-MM-Diffusion model. Although this dataset is already relatively substantial for traditional video prediction or regression tasks, it may still be insufficient for deep generative models (which include attention mechanisms, multi-scale convolutions in U-Net, etc.) to fully learn complex spatiotemporal features. In particular, California typically experiences many clear-sky days throughout the year; although cloudy conditions were separately filtered out, the variety of cloud types and cloud amounts in these samples remains limited. Consequently, the model may exhibit generalization errors when confronted with more extreme or rare cloud conditions.

If the framework is deployed at a new site, a practical strategy is to re-train or at least fine-tune the model using local data and adopt normalization schemes (e.g., scaling PV power by system capacity and/or clear-sky power) to reduce cross-system variability. A rigorous multi-site evaluation across different climates and PV technologies is an important direction for future work, but is beyond the scope of the present single-site dataset.

2. **Satellite imagery and scenarios without sky cameras:**

Operation without sky cameras. When sky images are unavailable, PV-MM-Diffusion can be deployed in the "PV→PV" mode (Section 2.5), using only historical PV measurements. This is a practical fallback for many real-world PV plants where cameras are not installed. However, Table 3 shows that the PV-only configuration yields a higher CRPS and a substantially worse Winkler score than the full multimodal setting, indicating degraded sharpness/reliability when explicit cloud-motion information is missing. In such cases, incorporating alternative exogenous inputs (e.g., NWP, irradiance sensors, or satellite nowcasting products) is expected to improve reliability and reduce the need for overly wide prediction intervals.

Potential use of satellite imagery. Replacing ground-based sky cameras with satellite imagery is conceptually feasible within our architecture: the image branch can ingest temporally aligned sequences of satellite patches after reprojection to a common grid and appropriate downsampling. Compared with fisheye sky cameras, satellite imagery typically offers a much larger field of view (beneficial for longer horizons and regional coverage) but has coarser spatial/temporal resolution, which may reduce minute-level forecast skill. Due to the lack of co-registered satellite–PV datasets within the current experimental setting, we do not conduct satellite-based experiments in this manuscript; therefore, the performance under satellite inputs remains to be validated in future work.

More generally, besides ground-based sky images, information from geostationary satellites, numerical weather prediction (NWP), and weather station measurements (e.g., wind speed, aerosols) can provide valuable priors for PV forecasting. Incorporating these heterogeneous sources may require dedicated encoder branches and cross-modal fusion modules to handle different spatiotemporal scales, which we leave for future work.

3. **Regarding computational overhead and improvements in model architecture:**

By employing the DPM-Solver++ accelerated sampling strategy, we substantially reduce the inference overhead of the reverse diffusion process; nevertheless, diffusion models still require iterative denoising and repeated sampling to form a predictive distribution. In our implementation, training the final model for 100,000 optimization steps takes **43.06 hours** on **8× NVIDIA RTX 3090 GPUs**. For inference, using **DPM-Solver++ with 60 solver steps**, generating a predictive distribution with **10 samples** takes **45.3895 s** on a single **RTX 3090** (per forecast instance, averaged over instances). The inference latency grows approximately linearly with the number of samples and solver steps; therefore, real-world deployments can trade off accuracy for speed by reducing the sampling budget (e.g., 10 or 5 samples) and/or solver steps, and can further reduce latency via multi-GPU parallel sampling across independent samples. Shortening inference latency in real-world applications thus requires a comprehensive consideration of both hardware and algorithms. Future improvements may include: (1) model pruning or lightweight architectures. Drawing on mobile network designs (e.g., MobileNet [59]) or lightweight video prediction architectures [60], one could reduce parameter counts as much as possible without compromising predictive accuracy; (2)

distributed parallelization or using more GPUs. In industrial-scale deployments, parallelization can be used to accelerate inference, though it is necessary to balance the cost of computational resources; (3) more efficient sampling strategies. Techniques such as multistep adaptive denoising [61] could further reduce the total number of reverse diffusion steps or enable segmented sampling, thereby controlling latency. One must also balance sampling quantity and forecasting reliability: as shown in Fig. 9, the improvement in CRPS/WS from increasing the number of samples gradually converges, and excessively increasing the sampling count does not necessarily bring significant benefits.

4. **On quantifying uncertainty for longer forecasting horizons and larger regional scales:**

   This study mainly focuses on ultra-short-term forecasting (at 15-minute intervals) for a single point (a single photovoltaic system). However, with the large-scale integration of distributed photovoltaic systems into the power grid, the need for minute-to-hour-level, region-scale probabilistic forecasting is gradually increasing [40]. The reason for selecting a 15-minute forecasting horizon in sky-image-based ultra-short-term forecasting is that ground-based sky cameras have a limited field of view. Nonetheless, the modeling strategy proposed in this paper is not confined to this time interval or a single site: by employing a sky imaging network [62] or incorporating satellite observational data [40], one could extend the forecasting scope from minutes to hours (for instance, 15–120 minutes) and cover multiple photovoltaic sites or larger grid areas, thereby providing more spatiotemporally consistent decision support for real-time scheduling and energy management.

6. Conclusion

This study confronts the rapid variability and uncertainty inherent in ultra-short-term photovoltaic (PV) power forecasting for high-renewable-penetration grids by proposing a unified multimodal diffusion model, *PV-MM-Diffusion*. Through two coupled denoising autoencoders, the model is capable of *simultaneously* generating future sky images and PV power sequences within a single end-to-end framework, while adaptively supporting heterogeneous input-output modalities. Large-scale experiments on real multi-cloud datasets show that the proposed model significantly outperforms multiple baselines in probabilistic metrics such as the Continuous Ranked Probability Score (CRPS) and the Winkler Score (WS). In extreme ramp-rate scenarios in particular, the model delivers more precise interval forecasts and superior risk characterisation, thereby overcoming the constraints of conventional “two-stage” methods in information exchange and uncertainty propagation. Moreover, a systematic investigation of unimodal and multimodal configurations corroborates the pronounced value of multimodal fusion in enhancing both forecasting robustness and accuracy. Considering the limitations and prospective improvements discussed herein, future work may extend the model to larger scales, incorporate more diverse meteorological data, and explore rapid-inference algorithms, ultimately providing resilient and high-quality uncertainty forecasts for broader renewable-energy scheduling and climate-modeling applications.

CRediT authorship contribution statement

Jing Huang: Writing – original draft, Project administration, Methodology, Formal analysis, Data curation. **Bopeng Shao:** Writing – review & editing, Writing – original draft, Validation, Software, Resources, Methodology, Investigation. **Yan Ke:** Writing – review & editing, Visualization, Supervision, Project administration. **Yuan Gao:** Writing – review & editing, Writing – original draft, Validation. **Stefano Mazzoni:** Writing – review & editing, Validation, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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